Cube Reversi Opening Book

A study guide to the named openings in Cube Reversi 1.35.1: the Classic 8×8 board and the 4×4×4, 6×6×6 and 8×8×8 cubes

How to use this guide

Every named opening that Cube Reversi 1.35.1 recognises is listed here with its move sequence, a diagram of the position after the last move, a short note on the idea, and the best replies. **Name Openings** is on by default, so the name of the opening you are playing appears in the top-left corner of the board. Untick it in the side panel, or press **O**, to hide the name; press **O** again to bring it back.

On the cube boards the Bot plays these book moves instantly. Hard always plays the best book move; Medium and Easy choose among the near-best ones, so you will meet several of the side lines in practice.

Reading the notation

Classic (8×8): a move is a column letter A–H and a row number 1–8, written as in the game's move list: `F-5`. Black always moves first; the four centre discs are D4 and E5 (White) and D5 and E4 (Black).

Cubes: a move has three parts, `column-row-layer`. The column is a letter from A, the row is a number from 1, and the layer is a letter from S. So `A-1-V` on the 4×4×4 cube is column A, row 1, layer V (the fourth layer); the layers are S, T, U, V on 4×4×4, S–X on 6×6×6 and S–Z on 8×8×8. This is exactly the notation used in the game's move-sequence box.

Start position on the cubes: the eight centre cells form a 2×2×2 block coloured like a 3D checkerboard: each centre cell is Black or White, and no two face-neighbours share a colour. Black moves first.

Reading the diagrams

Classic diagrams show the whole board with column letters across the top and row numbers down the side. Cube diagrams show the cube **layer by layer**: one small board for each layer (Layer S, Layer T, …), each drawn the same way as a Classic board. Stack the layers on top of each other in order and you have the cube.

Black discs are solid black, White discs are white, and the **red dot** marks the last move played. Under each opening the line “discs 6–3” gives the Black–White disc count and says who is to move.

Symmetry: rotated and mirrored openings are the same opening

The board has no preferred side, so a position that is merely rotated or reflected is the same position. The game knows this: it recognises each opening in every symmetric form, and a different move order that reaches the same position (a transposition) gets the same name.

On the Classic board four symmetries keep the start position (the identity, a half turn and the two diagonal mirrors). The book lists each opening in the standard F-5 orientation, but you may meet it as D-3, C-4 or E-6: the same first move turned or mirrored.

On the cubes there are 48 ways to turn or mirror a cube. 24 of them leave the start position unchanged (12 rotations and 12 reflections); the other 24 turn it into the colour-swapped start. The game uses all 48 together with swapping the colours, so an opening is recognised whichever corner, face or direction you play it in. For example, on 4×4×4 Black's first move A-1-V can just as well be played as A-4-S, D-1-S

or D-4-V; all four are the *Vertex Opening*, and all 12 non-corner first moves are the *Facet Opening*.

What the scores mean

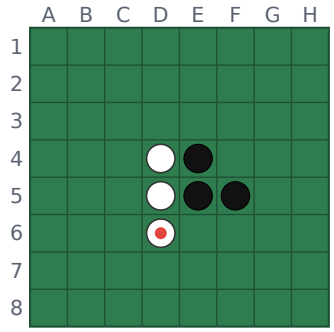
The cube openings were analysed by the game's own engine (about 10 seconds of search per position on 4×4×4, 20 seconds on the larger cubes). “Best replies” are the moves the engine rated highest for the side to move, in the orientation shown in the diagram. They are strong practical choices, not a mathematical solution of the game.

Part 1 — Classic openings (8×8 board)

These are the established Othello openings the game already knows. In 1.35.1 the catalogue was tidied to 47 named lines, each with a unique name and a unique move sequence (duplicate entries were removed, and Buffalo – Kenichi, which had exactly the same moves as Buffalo, is now listed only as Buffalo). In 2.0.9 Aubrey / Tanaka was corrected to F-5 D-6 C-3 D-3 C-4 B-3 and Nicolet was added, giving 48 lines. Black's first move is always shown as F-5; White then chooses the **Perpendicular** (D-6), **Diagonal** (F-6) or **Parallel** (F-4) reply, and the named openings grow from there. “Book continuations” lists the next moves that lead to longer named lines in the game's catalogue.

Perpendicular

`F-5 D-6` (2 moves; discs 3–3; Black to move)

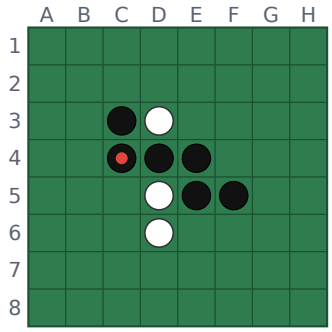


White replies at right angles to Black's first move (F-5 D-6). The most popular reply and the root of the Tiger, Rose, Cat, Horse and related openings.

Book continuations: `C-3` (Tiger, Tiger - Aubrey / Tanaka, Tiger - Rose-Bill, Tiger - Nicolet, +8 more); `C-5` (Horse, Rose, Rose - Ralle, Rose - Inoue, +5 more); `C-4` (Cat, Cat - Berner, Cat - Sakaguchi, Cat - Italian, +2 more)

Tiger

`F-5 D-6 C-3 D-3 C-4` (5 moves; discs 6–3; White to move)

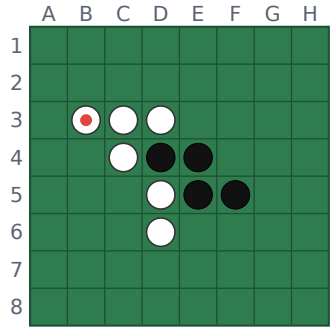


Black plays C-3 on move 3 (F5 D6 C3 D3 C4). The most widely played Perpendicular line; it leads to sharp, well-studied variations.

Book continuations: `B-3` (Aubrey / Tanaka); `F-4` (Rose-Bill, Nicolet, Tamenori, Leader's Tiger, +5 more)

Tiger — Aubrey / Tanaka

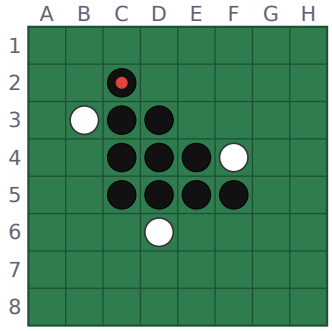
`F-5 D-6 C-3 D-3 C-4 B-3` (6 moves; discs 4–6; Black to move)



A named branch of the Tiger, defined by its last move B-3.

Tiger — Rose-Bill

`F-5 D-6 C-3 D-3 C-4 F-4 C-5 B-3 C-2` (9 moves; discs 10–3; White to move)

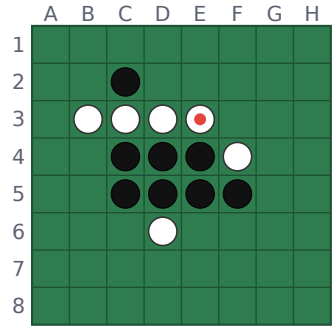


A named branch of the Tiger, defined by its last move C-2.

Book continuations: `E-3` (Nicolet); `E-6` (Tamenori)

Tiger — Nicolet

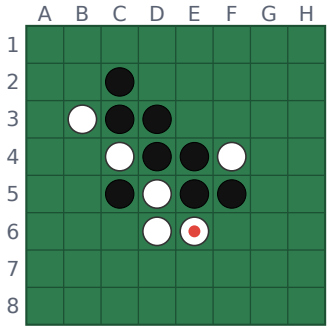
F-5 D-6 C-3 D-3 C-4 F-4 C-5 B-3 C-2 E-3 (10 moves; discs 8–6; Black to move)



A named branch of the Tiger, defined by its last move E-3.

Tiger — Tamenori

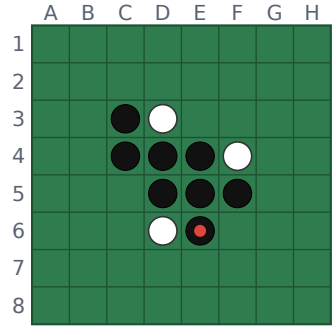
F-5 D-6 C-3 D-3 C-4 F-4 C-5 B-3 C-2 E-6 (10 moves; discs 8–6; Black to move)



A named branch of the Tiger, defined by its last move E-6.

Tiger — Leader's Tiger

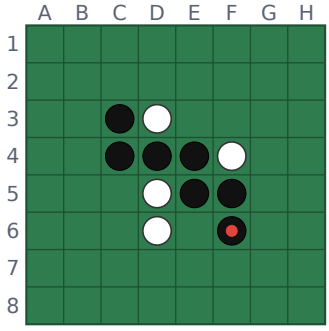
F-5 D-6 C-3 D-3 C-4 F-4 E-6 (7 moves; discs 8–3; White to move)



A named branch of the Tiger, defined by its last move E-6.

Tiger — Stephenson

F-5 D-6 C-3 D-3 C-4 F-4 F-6 (7 moves; discs 7–4; White to move)

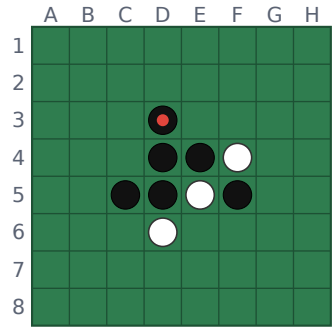


A named branch of the Tiger, defined by its last move F-6.

Book continuations: B-4 (Kung); F-3 (Comp'Oth); G-5 (No-Kung)

Horse

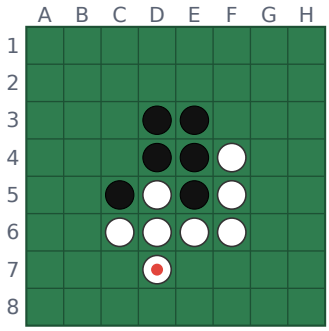
F-5 D-6 C-5 F-4 D-3 (5 moves; discs 6–3; White to move)



F5 D6 C5 F4 D3 — Black leaves the Rose path on move 5. Less common but sound.

Rose

F-5 D-6 C-5 F-4 E-3 C-6 D-3 F-6 E-6 D-7 (10 moves; discs 6–8; Black to move)

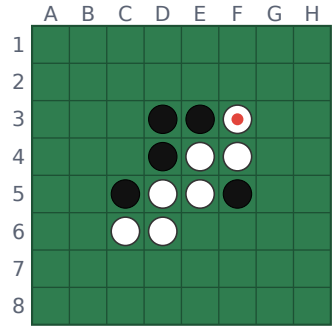


F5 D6 C5 F4 E3 C6 D3 F6 E6 D7 — one of the most studied main lines in Othello, often considered best play for both sides for many moves.

Book continuations: G-3 (Rose-Birth, Rose-birdie)

Rose — Ralle

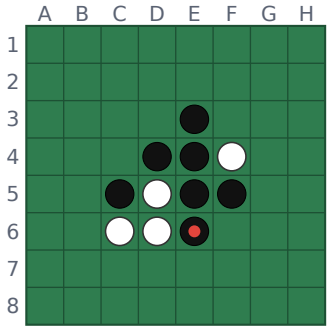
F-5 D-6 C-5 F-4 E-3 C-6 D-3 F-3 (8 moves; discs 5–7; Black to move)



A named branch of the Rose, defined by its last move F-3.

Rose — Inoue

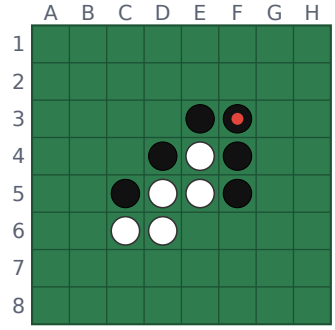
F-5 D-6 C-5 F-4 E-3 C-6 E-6 (7 moves; discs 7–4; White to move)



A named branch of the Rose, defined by its last move E-6.

Rose — Shaman / Danish

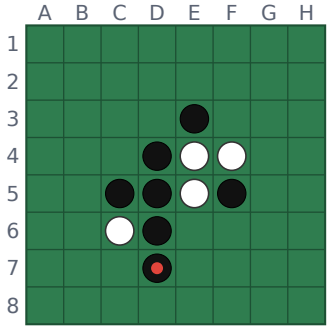
F-5 D-6 C-5 F-4 E-3 C-6 F-3 (7 moves; discs 6–5; White to move)



A named branch of the Rose, defined by its last move F-3.

Rose — Bhagat

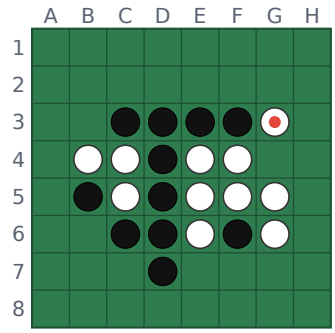
F-5 D-6 C-5 F-4 E-3 C-6 D-7 (7 moves; discs 7–4; White to move)



A named branch of the Rose, defined by its last move D-7.

Tiger — Mainline Tiger

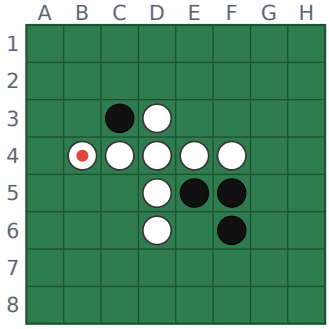
F-5 D-6 C-3 D-3 C-4 F-4 C-5 B-4 B-5 C-6 F-3 E-6 E-3 G-6 F-6 G-5 D-7 G-3 (18 moves; discs 11–11; Black to move)



The long main line of the Tiger: 18 moves deep in the game's catalogue, a good line to memorise move by move.

Tiger — Kung

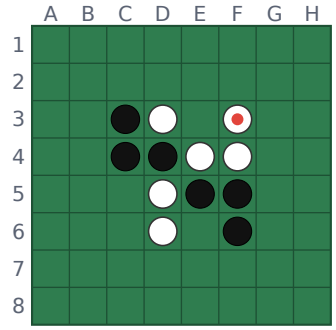
F-5 D-6 C-3 D-3 C-4 F-4 F-6 B-4 (8 moves; discs 4–8; Black to move)



A named branch of the Tiger, defined by its last move B-4.

Tiger — Comp'Oth

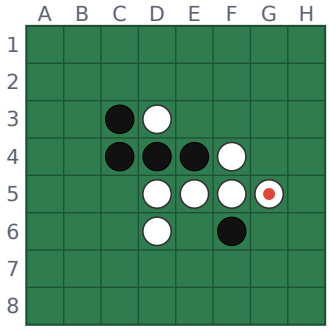
F-5 D-6 C-3 D-3 C-4 F-4 F-6 F-3 (8 moves; discs 6–6; Black to move)



A named branch of the Tiger, defined by its last move F-3.

Tiger — No-Kung

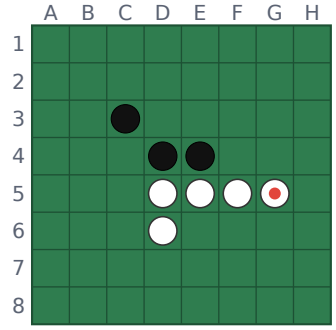
F-5 D-6 C-3 D-3 C-4 F-4 F-6 G-5 (8 moves; discs 5–7; Black to move)



A named branch of the Tiger, defined by its last move G-5.

Tiger — Ganglion

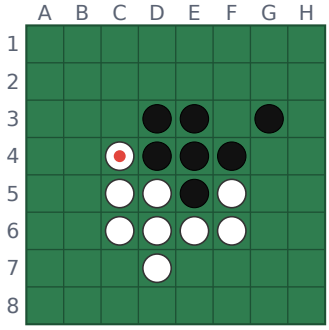
F-5 D-6 C-3 G-5 (4 moves; discs 3–5; Black to move)



White leaves the Tiger early with G-5 on move 4.

Rose — Rose-Birth

F-5 D-6 C-5 F-4 E-3 C-6 D-3 F-6 E-6 D-7 G-3 C-4 (12 moves; discs 7–9; Black to move)

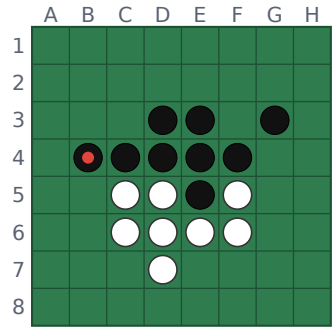


The Rose continued with G-3 and C-4 — the full Rose line extended by two more moves.

Book continuations: B-4 (Rose-birdie)

Rose — Rose-birdie

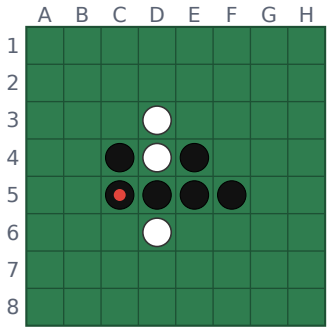
F-5 D-6 C-5 F-4 E-3 C-6 D-3 F-6 E-6 D-7 G-3 C-4 B-4 (13 moves; discs 9–8; White to move)



The Rose-Birth continued with B-4.

Cat

F-5 D-6 C-4 D-3 C-5 (5 moves; discs 6–3; White to move)

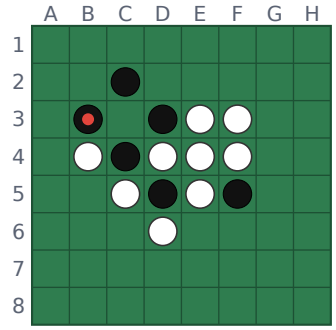


F5 D6 C4 D3 C5 — Black's third move C-4 gives a quieter, more positional game than the Tiger.

Book continuations: F-4 (Berner, Sakaguchi)

Cat — Berner

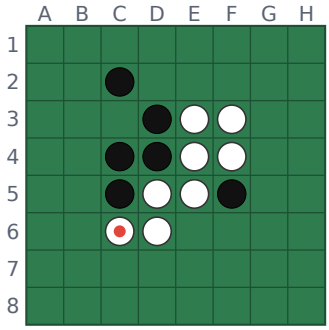
F-5 D-6 C-4 D-3 C-5 F-4 E-3 F-3 C-2 B-4 B-3 (11 moves; discs 6–9; White to move)



A named branch of the Cat, defined by its last move B-3.

Cat — Sakaguchi

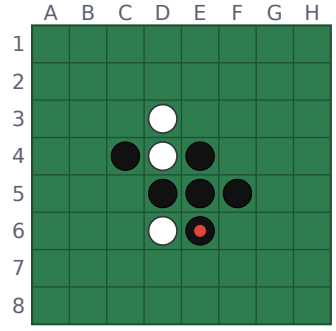
F-5 D-6 C-4 D-3 C-5 F-4 E-3 F-3 C-2 C-6 (10 moves; discs 6–8; Black to move)



A named branch of the Cat, defined by its last move C-6.

Cat — Italian

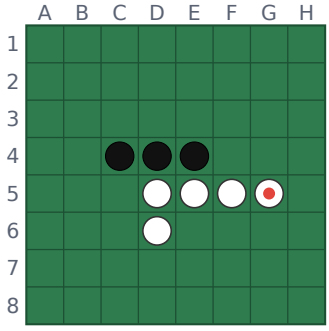
F-5 D-6 C-4 D-3 E-6 (5 moves; discs 6–3; White to move)



A named branch of the Cat, defined by its last move E-6.

Cat — No-Cat

F-5 D-6 C-4 G-5 (4 moves; discs 3–5; Black to move)

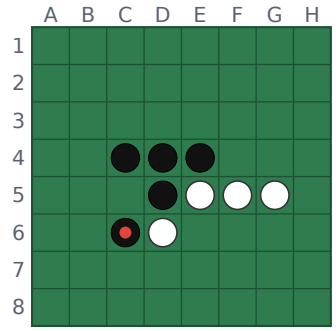


White avoids the Cat with G-5 on move 4.

Book continuations: C-6 (Swallow)

Cat — Swallow

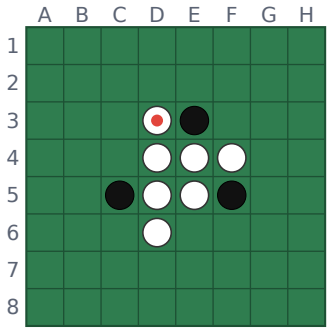
F-5 D-6 C-4 G-5 C-6 (5 moves; discs 5–4; White to move)



A named branch of the Cat, defined by its last move C-6.

Rose — Mimura

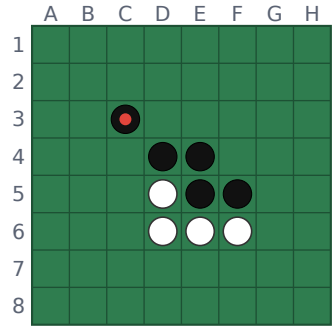
F-5 D-6 C-5 F-4 E-3 D-3 (6 moves; discs 3–7; Black to move)



A named branch of the Rose, defined by its last move D-3.

Buffalo

`F-5 F-6 E-6 D-6 C-3` (5 moves; discs 5–4; White to move)

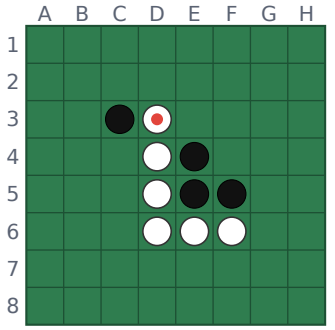


A Diagonal-opening line (F5 F6 E6 D6 C3). Black plays aggressively toward the C-3 area.

Book continuations: `D-3` (Hokuriku Buffalo); `F-4` (Tanida Buffalo); `G-4` (Maruoka Buffalo)

Buffalo — Hokuriku Buffalo

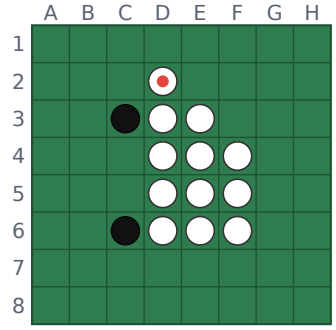
`F-5 F-6 E-6 D-6 C-3 D-3` (6 moves; discs 4–6; Black to move)



A named branch of the Buffalo, defined by its last move D-3.

Buffalo — Tanida Buffalo

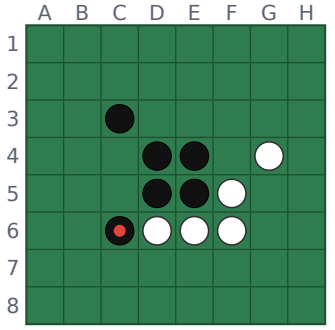
`F-5 F-6 E-6 D-6 C-3 F-4 C-6 D-3 E-3 D-2` (10 moves; discs 2–12; Black to move)



A named branch of the Buffalo, defined by its last move D-2.

Buffalo — Maruoka Buffalo

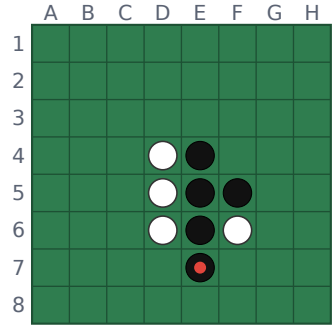
`F-5 F-6 E-6 D-6 C-3 G-4 C-6` (7 moves; discs 6–5; White to move)



A named branch of the Buffalo, defined by its last move C-6.

Heath

`F-5 F-6 E-6 D-6 E-7` (5 moves; discs 5–4; White to move)

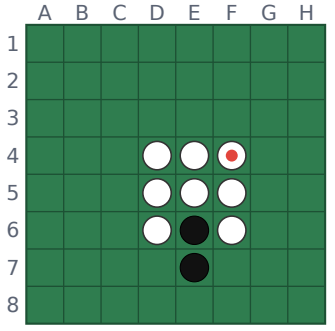


F5 F6 E6 D6 E7 — a Diagonal line in which Black plays E-7 on move 5; it has several named branches.

Book continuations: `F-4` (Heath-Chimney / Mass-Turning); `G-5` (Heath-Bat, Iwasaki, Mimura II)

Heath — Heath-Chimney / Mass-Turning

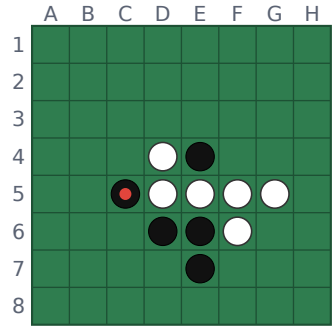
`F-5 F-6 E-6 D-6 E-7 F-4` (6 moves; discs 2–8; Black to move)



A named branch of the Heath, defined by its last move F-4.

Heath — Heath-Bat

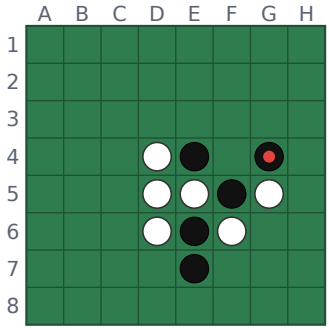
F-5 F-6 E-6 D-6 E-7 G-5 C-5 (7 moves; discs 5–6; White to move)



A named branch of the Heath, defined by its last move C-5.

Heath — Iwasaki

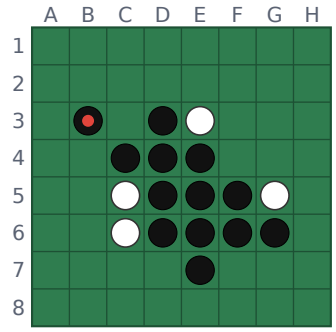
F-5 F-6 E-6 D-6 E-7 G-5 G-4 (7 moves; discs 5–6; White to move)



A named branch of the Heath, defined by its last move G-4.

Heath — Mimura II

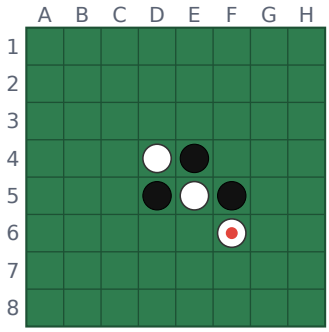
F-5 F-6 E-6 D-6 E-7 G-5 G-6 E-3 C-5 C-6 D-3 C-4 B-3 (13 moves; discs 13–4; White to move)



A named branch of the Heath, defined by its last move B-3.

Diagonal

F-5 F-6 (2 moves; discs 3–3; Black to move)

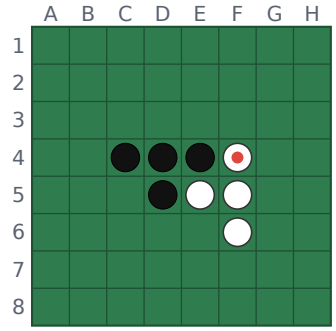


White replies diagonally (F5 F6). A strong alternative to the Perpendicular, with the Buffalo, Heath, Cow and other lines.

Book continuations: E-6 (Buffalo, Buffalo - Hokuriku Buffalo, Buffalo - Tanida Buffalo, Buffalo - Maruoka Buffalo, +11 more); C-4 (Semi-Wing); D-3 (Wing)

Diagonal — Semi-Wing

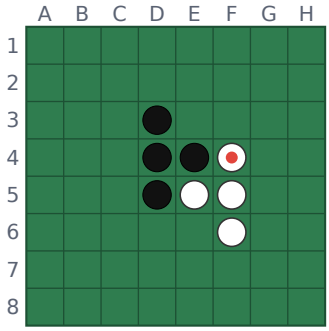
F-5 F-6 C-4 F-4 (4 moves; discs 4–4; Black to move)



A Diagonal line with C-4 and F-4 on moves 3–4.

Diagonal — Wing

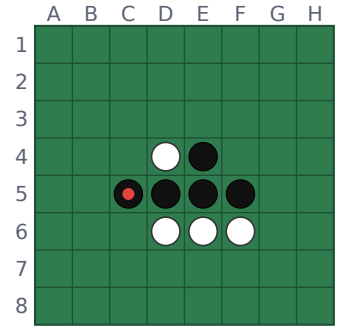
F-5 F-6 D-3 F-4 (4 moves; discs 4–4; Black to move)



A Diagonal line with D-3 and F-4 on moves 3–4.

Cow

`F-5 F-6 E-6 D-6 C-5` (5 moves; discs 5–4; White to move)

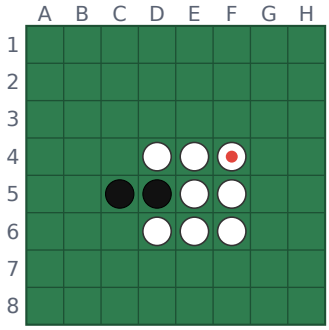


F5 F6 E6 D6 C5 — a solid Diagonal line; its continuation F-4 is the Chimney.

Book continuations: F-4 (Chimney)

Chimney

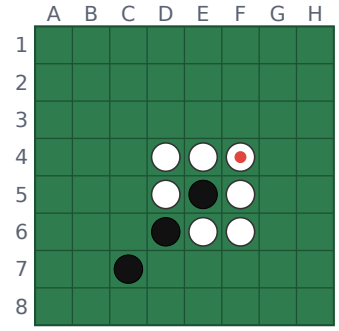
`F-5 F-6 E-6 D-6 C-5 F-4` (6 moves; discs 2–8; Black to move)



F5 F6 E6 D6 C5 F4 — the Cow continued with F-4.

Lollipop

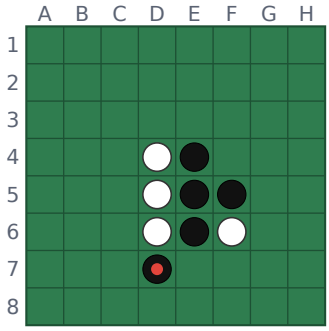
`F-5 F-6 E-6 D-6 C-7 F-4` (6 moves; discs 3–7; Black to move)



F5 F6 E6 D6 C7 F4 — an unusual Diagonal sideline.

Raccoon Dog

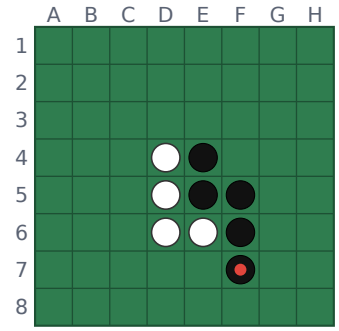
`F-5 F-6 E-6 D-6 D-7` (5 moves; discs 5–4; White to move)



F5 F6 E6 D6 D7 — a Diagonal sideline with Black's fifth move on D-7.

Snake / Peasant

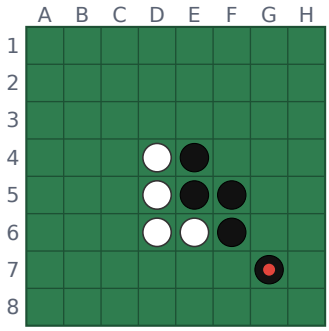
`F-5 F-6 E-6 D-6 F-7` (5 moves; discs 5–4; White to move)



F5 F6 E6 D6 F7 — a Diagonal sideline.

X-square

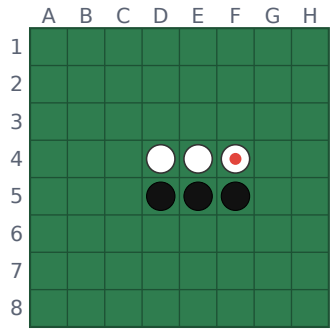
`F-5 F-6 E-6 D-6 G-7` (5 moves; discs 5–4; White to move)



F5 F6 E6 D6 G7 — Black plays next to the corner (an 'X-square'), usually a risky idea because it can give the corner away.

Parallel

`F-5 F-4` (2 moves; discs 3–3; Black to move)

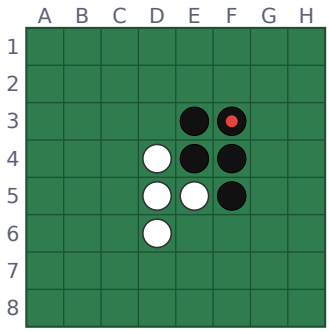


White replies parallel to Black's first move (F5 F4). Generally regarded as the weakest of White's three replies.

Book continuations: E-3 (Mouse)

Parallel — Mouse

`F-5 F-4 E-3 D-6 F-3` (5 moves; discs 5–4; White to move)



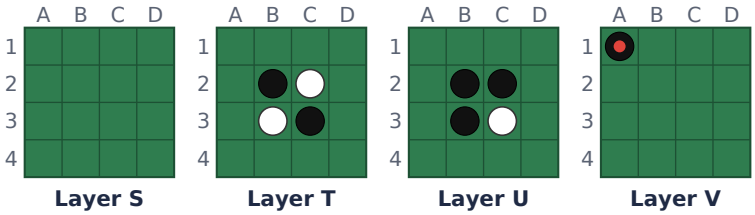
The best-known continuation of the Parallel opening.

Part 2 — 4×4×4 cube openings

On the small cube the corners are within reach from the very first move, and the opening is a race for them: a corner can never be flipped. The engine prefers taking a corner at once (Peter Alfeld's 3D Reversi pages already point out that this is possible). There are only two different first moves up to symmetry: the Vertex Opening (a corner) and the Facet Opening (any other first move). Each White reply opens its own family. The book has 252 analysed positions. The deepest named line is 11 plies long (Snake Eyes — Hypercube), and from its last position the book still supplies the next reply.

Vertex Opening

`A-1-V` (1 move; discs 6–3; White to move)

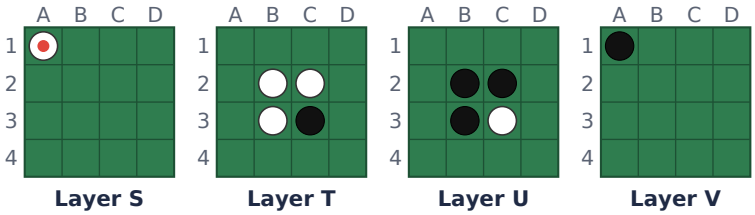


Black takes a corner on the very first move — something the flat board never allows. The engine's choice: a corner can never be flipped.

Best replies (White): `A-1-S`

Snake Eyes

`A-1-V A-1-S` (2 moves; discs 5–5; Black to move)

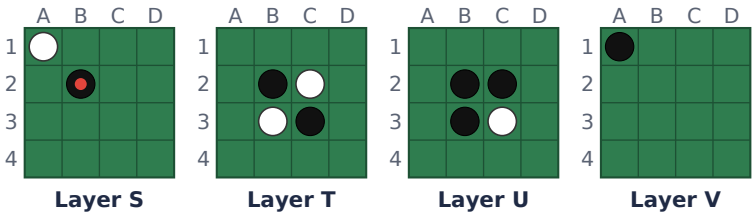


White answers a corner with a corner of its own on the same edge. The engine's best reply: both sides now own an unflippable anchor.

Best replies (Black): `B-2-S`, `A-4-S`, `A-2-S`

Snake Eyes — Rubik's Line

`A-1-V A-1-S B-2-S` (3 moves; discs 7–4; White to move)

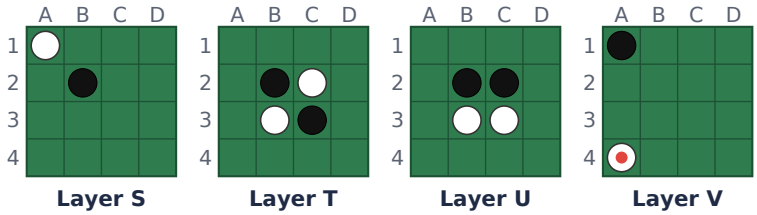


Black twists into the bottom face next to White's corner, building a wall of stable discs; the main line of the book.

Best replies (White): `A-4-V`, `A-2-T`

Snake Eyes — Boxcars

`A-1-V A-1-S B-2-S A-4-V` (4 moves; discs 6–6; Black to move)

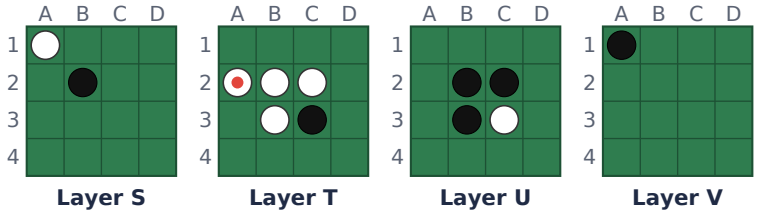


White grabs a third corner straight away — two corners each would be 'boxcars'. Black's best follow-up is B-4-U.

Best replies (Black): `B-4-U`, `A-4-S`

Snake Eyes — Möbius Strip

`A-1-V A-1-S B-2-S A-2-T` (4 moves; discs 6–6; Black to move)

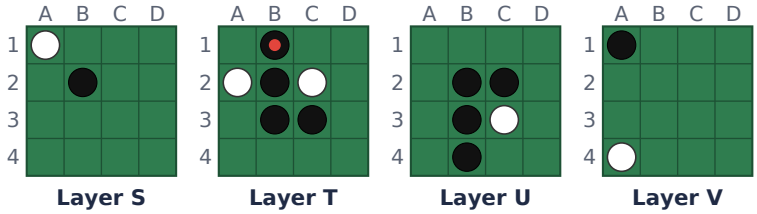


White declines the corner race and slides along the edge with A-2-T; Black usually replies A-4-S or A-2-S.

Best replies (Black): `A-4-S`, `A-2-S`, `D-1-S`

Snake Eyes — Menger Sponge

`A-1-V A-1-S B-2-S A-4-V B-4-U A-2-T B-1-T` (7 moves; discs 10–5; White to move)

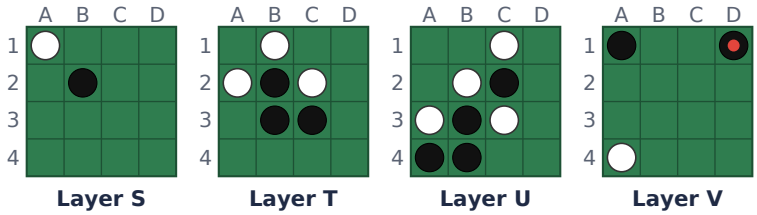


The main line digs holes into the A-face edges like a Menger sponge; each side keeps its corner groups connected.

Best replies (White): `A-3-U`, `C-4-V`

Snake Eyes — Hypercube

`A-1-V A-1-S B-2-S A-4-V B-4-U A-2-T B-1-T A-3-U A-4-U C-1-U D-1-V` (11 moves; discs 10–9; White to move)

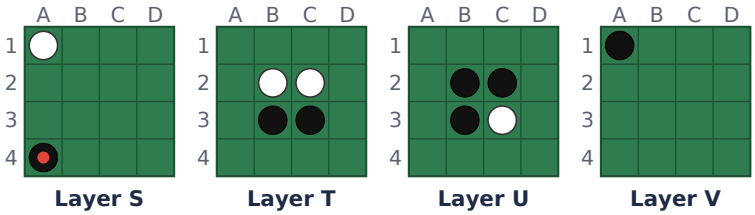


The deepest main line in the book (11 plies): Black finally takes the D-1-V corner. Best continuation D-4-S or D-2-U.

Best replies (White): `D-4-S`, `D-2-U`, `C-2-V`

Snake Eyes — Trinity

`A-1-V` `A-1-S` `A-4-S` (3 moves; discs 7–4; White to move)

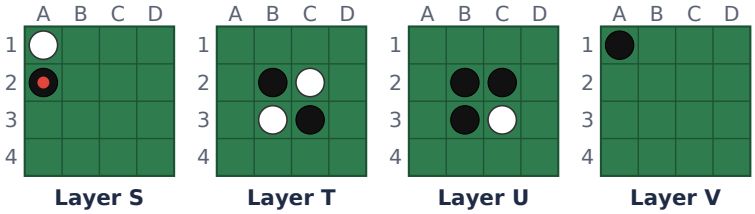


Black takes a second corner at once. White must answer A-4-V; tempting, but slightly worse for Black than Rubik's Line.

Best replies (White): `A-4-V`

Snake Eyes — Octahedron

`A-1-V` `A-1-S` `A-2-S` (3 moves; discs 7–4; White to move)

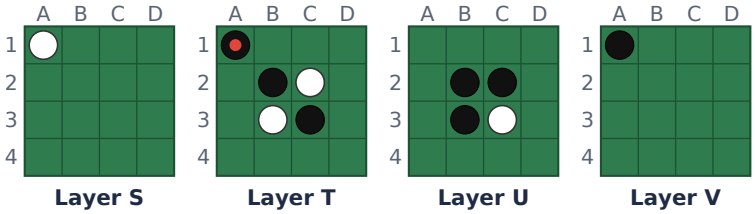


Black hugs White's corner from the S face; White's best is the A-4-V corner.

Best replies (White): `A-4-V`

Snake Eyes — Dodecahedron

`A-1-V` `A-1-S` `A-1-T` (3 moves; discs 7–4; White to move)

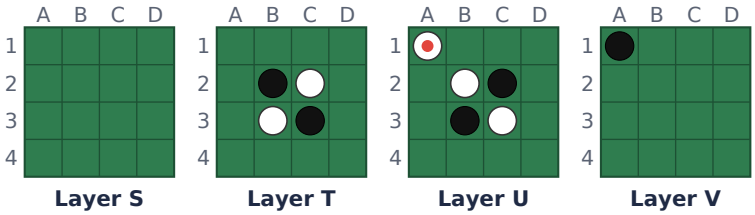


Black lines up on White's edge; White answers A-4-V or A-2-T.

Best replies (White): `A-4-V`, `A-2-T`

Edge Walker

`A-1-V` `A-1-U` (2 moves; discs 5–5; Black to move)

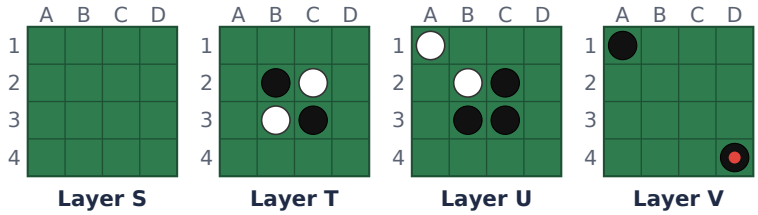


White steps onto the edge next to Black's corner instead of taking a corner. Safe-looking but passive.

Best replies (Black): `D-4-V`, `A-4-S`

Edge Walker — Far Corner

`A-1-V A-1-U D-4-V` (3 moves; discs 7–4; White to move)

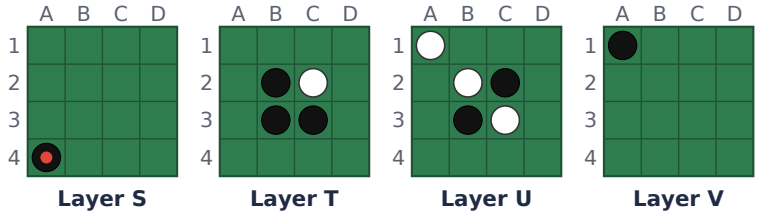


Black answers by taking the D-4-V corner across the cube; White replies A-4-V or D-4-S.

Best replies (White): `A-4-V`, `D-4-S`

Edge Walker — Tightrope

`A-1-V A-1-U A-4-S` (3 moves; discs 7–4; White to move)

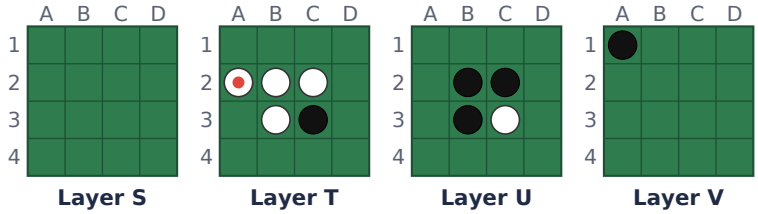


Black walks the tightrope along the A-face edge to the A-4-S corner; White answers A-1-S, taking a corner back.

Best replies (White): `A-1-S`, `A-4-V`, `A-3-S`

Wormhole

`A-1-V A-2-T` (2 moves; discs 5–5; Black to move)

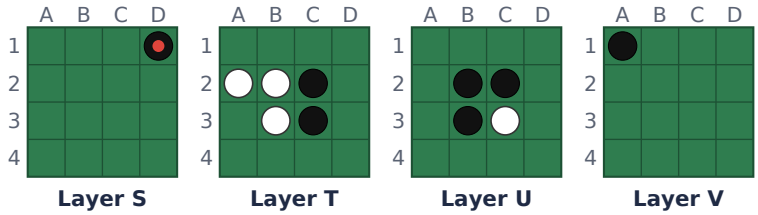


White tunnels into the middle of the A face. Black's clear answer is the D-1-S corner.

Best replies (Black): `D-1-S`

Wormhole — Singularity

`A-1-V A-2-T D-1-S` (3 moves; discs 7–4; White to move)

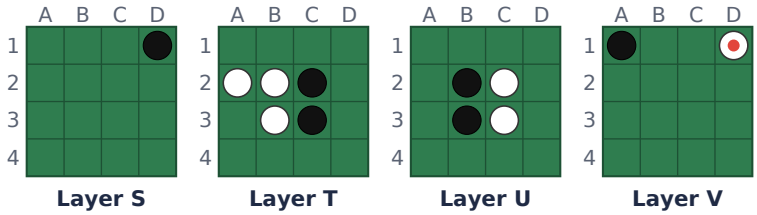


Black's best reply: a second corner. White usually follows with D-1-V or D-2-V.

Best replies (White): `D-1-V`, `D-2-V`

Wormhole — Tidal Lock

`A-1-V A-2-T D-1-S D-1-V` (4 moves; discs 6–6; Black to move)

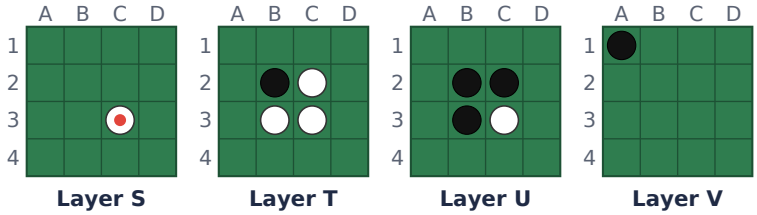


White locks the D-1 edge with the neighbouring corner; Black's best is C-2-V.

Best replies (Black): `C-2-V`, `D-2-U`, `D-1-U`

Black Hole

`A-1-V C-3-S` (2 moves; discs 5–5; Black to move)

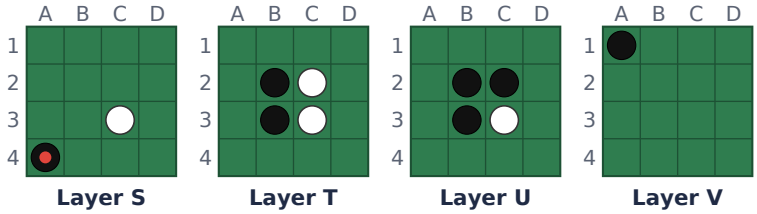


A dubious reply for White: it lets Black reach corners easily, and the evaluation falls off a cliff.

Best replies (Black): `A-4-S`, `D-4-V`

Black Hole — Accretion Disc

`A-1-V C-3-S A-4-S` (3 moves; discs 7–4; White to move)

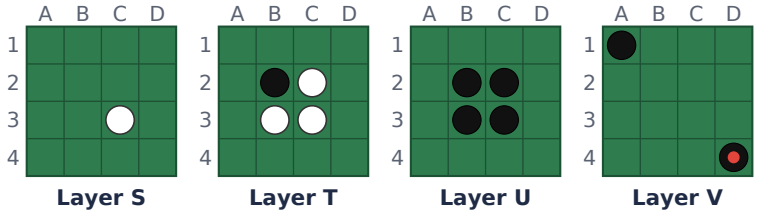


Black takes the A-4-S corner and builds a disc of stable discs around it.

Best replies (White): `A-4-V`, `A-4-T`

Black Hole — Gravity Well

`A-1-V C-3-S D-4-V` (3 moves; discs 7–4; White to move)

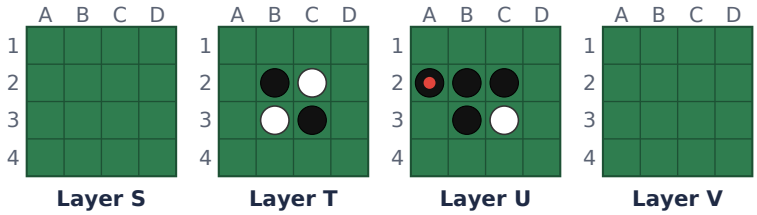


Black takes the D-4-V corner instead — just as strong.

Best replies (White): `A-4-V`, `C-4-V`

Facet Opening

`A-2-U` (1 move; discs 6–3; White to move)

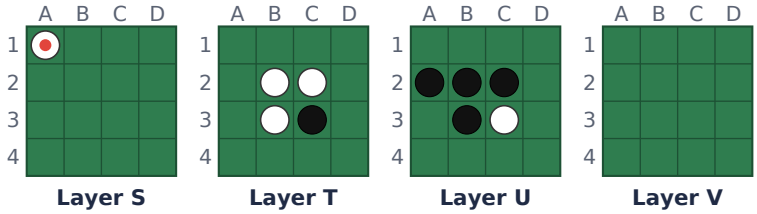


Black plays on a face edge instead of the corner. Playable, but it hands White the first corner; the engine rates it clearly below the Vertex Opening.

Best replies (White): `A-1-S`, `D-1-V`

Stolen Vertex

`A-2-U A-1-S` (2 moves; discs 5–5; Black to move)

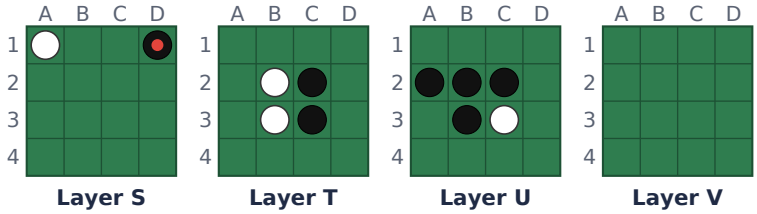


After the Facet Opening White simply steals the first corner — joint-best with Antipodes.

Best replies (Black): `D-1-S`, `A-4-S`, `A-1-T`

Stolen Vertex — Euclid

`A-2-U A-1-S D-1-S` (3 moves; discs 7–4; White to move)

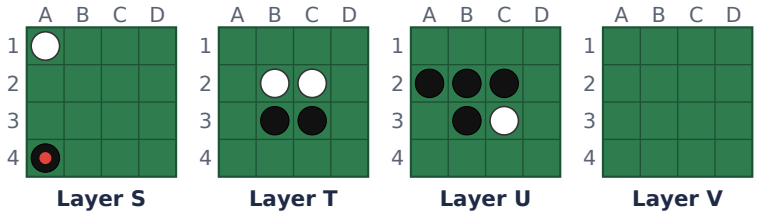


Black takes the neighbouring corner on the same edge; White answers D-1-V. The best line for Black.

Best replies (White): `D-1-V`

Stolen Vertex — Archimedes

`A-2-U A-1-S A-4-S` (3 moves; discs 7–4; White to move)

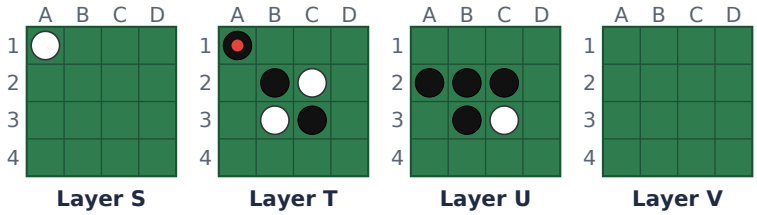


Black grabs the A-4-S corner; White answers A-4-V.

Best replies (White): `A-4-V`

Stolen Vertex — Pythagoras

`A-2-U A-1-S A-1-T` (3 moves; discs 7–4; White to move)

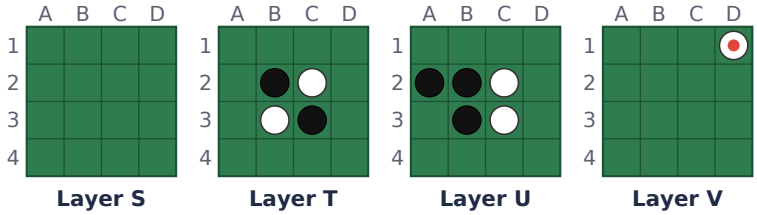


Black crowds White's corner; White replies A-4-V or A-2-T with a big advantage.

Best replies (White): `A-4-V`, `A-2-T`

Antipodes

`A-2-U D-1-V` (2 moves; discs 5–5; Black to move)

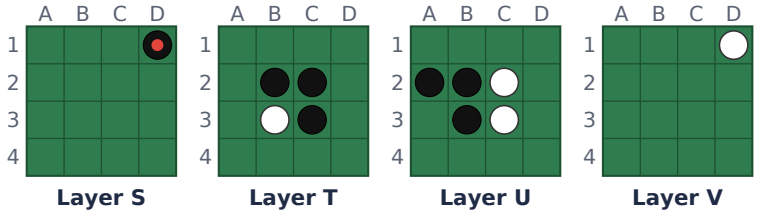


White takes the corner on the opposite side of the cube from Black's first disc. Equal-best with Stolen Vertex.

Best replies (Black): `D-1-S`, `C-1-V`, `D-2-U`

Antipodes — Equator

`A-2-U D-1-V D-1-S` (3 moves; discs 7–4; White to move)

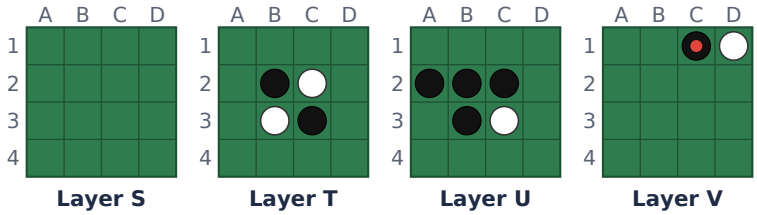


Black takes the neighbouring corner on White's edge; White answers B-1-T.

Best replies (White): `B-1-T`, `D-1-T`, `C-1-S`

Antipodes — Meridian

`A-2-U D-1-V C-1-V` (3 moves; discs 7–4; White to move)

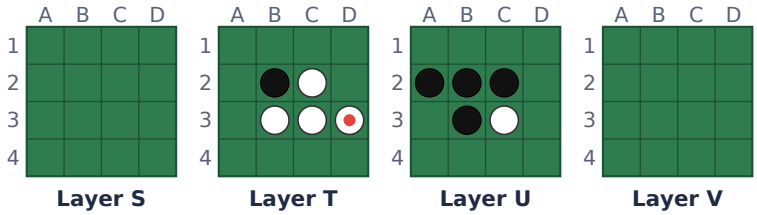


Black crowds White's corner; White answers A-1-S, taking a second corner.

Best replies (White): `A-1-S`, `C-1-U`

Tesseract

`A-2-U D-3-T` (2 moves; discs 5–5; Black to move)

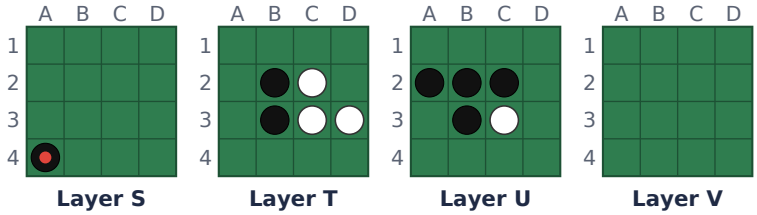


White plays deep into the D face; Black answers with a corner (D-1-S, D-4-S or A-4-S).

Best replies (Black): `D-1-S`, `D-4-S`, `A-4-S`

Tesseract — Unfolded

`A-2-U D-3-T A-4-S` (3 moves; discs 7–4; White to move)

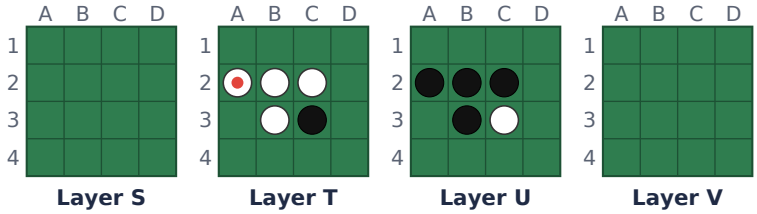


Black takes the A-4-S corner; White answers A-1-S or A-1-V.

Best replies (White): `A-1-S`, `A-1-V`, `A-3-T`

Escher Stairs

`A-2-U A-2-T` (2 moves; discs 5–5; Black to move)

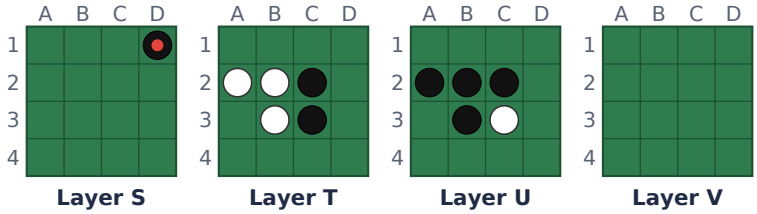


White climbs one step along the A face; Black's best is the D-1-S or A-4-S corner.

Best replies (Black): `D-1-S`, `A-4-S`

Escher Stairs — Waterfall

`A-2-U A-2-T D-1-S` (3 moves; discs 7–4; White to move)

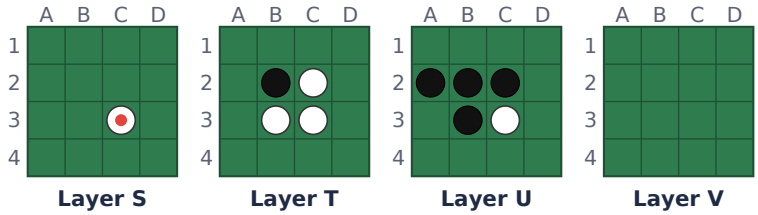


Black takes D-1-S; White answers D-1-V or D-2-V.

Best replies (White): `D-1-V`, `D-2-V`

Quasar

`A-2-U C-3-S` (2 moves; discs 5–5; Black to move)

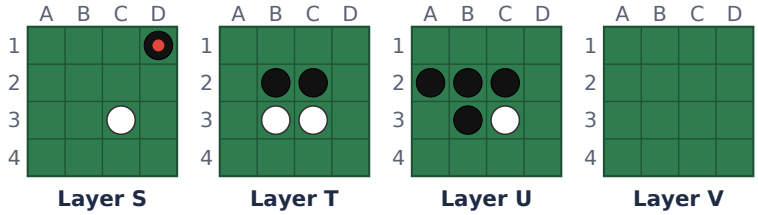


White's disc on the S face leaves corners open; Black takes D-1-S.

Best replies (Black): `D-1-S`

Quasar — Jet

`A-2-U C-3-S D-1-S` (3 moves; discs 7–4; White to move)

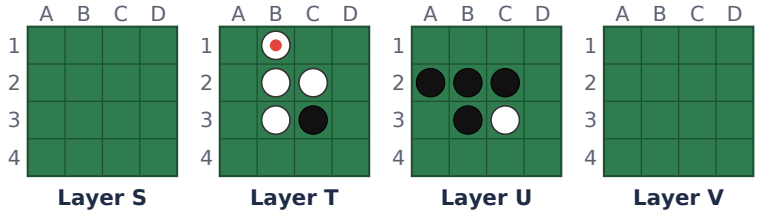


Black's best reply; White usually takes the A-1-S corner.

Best replies (White): `A-1-S`

Kepler

`A-2-U B-1-T` (2 moves; discs 5–5; Black to move)

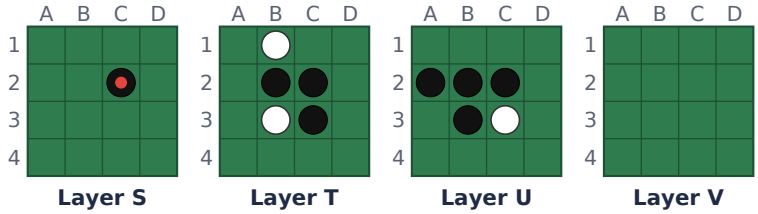


White plays on the bottom edge; Black answers C-2-S or the A-4-S corner.

Best replies (Black): `C-2-S`, `A-4-S`

Kepler — Ellipse

`A-2-U B-1-T C-2-S` (3 moves; discs 8–3; White to move)

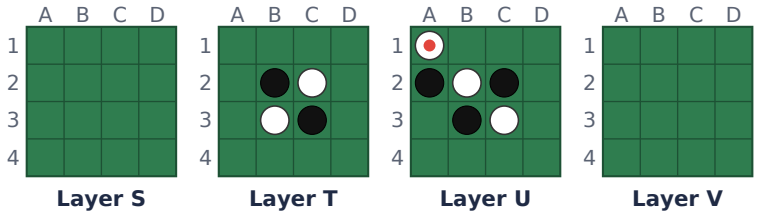


Black's best reply; White answers by taking the A-1-S corner.

Best replies (White): `A-1-S`

Pulsar

`A-2-U A-1-U` (2 moves; discs 5–5; Black to move)

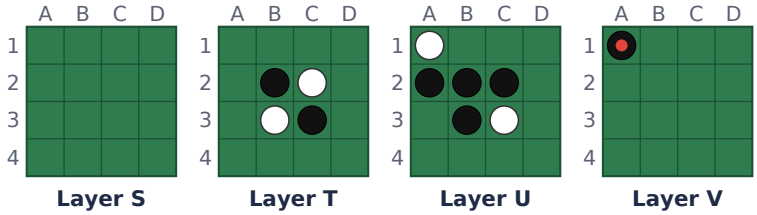


White steps next to the A-1-V corner — and Black takes it at once.

Best replies (Black): `A-1-V`

Pulsar — Lighthouse

`A-2-U A-1-U A-1-V` (3 moves; discs 7–4; White to move)

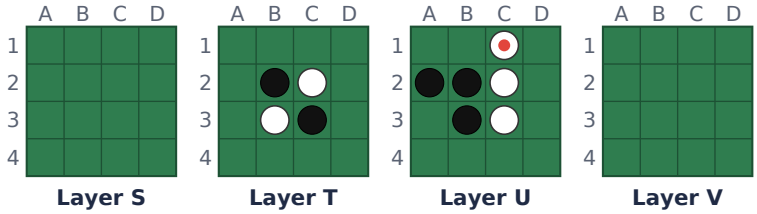


Black's best: the corner. White answers A-4-V.

Best replies (White): `A-4-V`

Orbit

`A-2-U C-1-U` (2 moves; discs 5–5; Black to move)

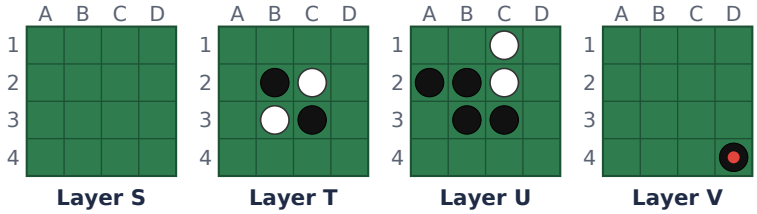


White circles to the bottom face; Black answers D-4-V.

Best replies (Black): `D-4-V`

Orbit — Slingshot

`A-2-U C-1-U D-4-V` (3 moves; discs 7–4; White to move)

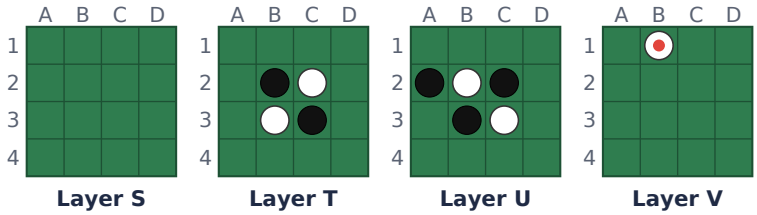


Black takes the D-4-V corner; White replies A-4-V, A-4-U or B-3-V.

Best replies (White): `A-4-V`, `A-4-U`, `B-3-V`

Platonic Defence

`A-2-U B-1-V` (2 moves; discs 5–5; Black to move)

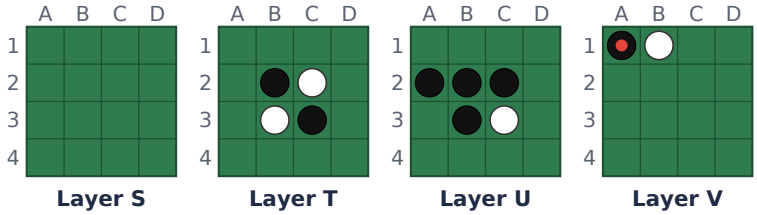


White sits next to the A-1-V corner; Black takes it.

Best replies (Black): `A-1-V`

Platonic Defence — Icosahedron

`A-2-U B-1-V A-1-V` (3 moves; discs 7–4; White to move)



Black's best reply: the corner. White answers A-1-S or D-1-V.

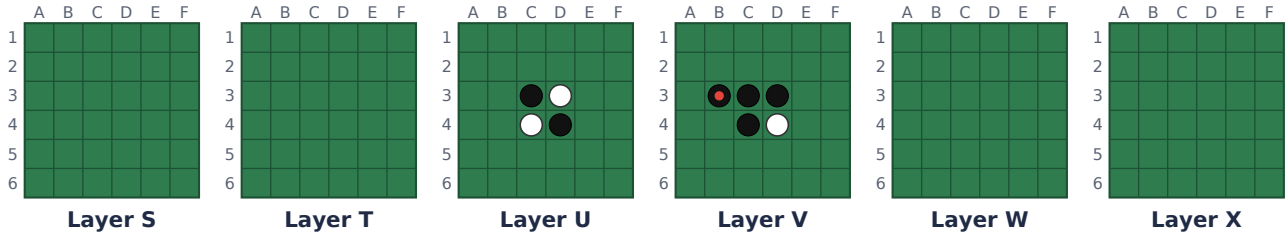
Best replies (White): `A-1-S`, `D-1-V`

Part 3 — 6×6×6 cube openings

On 6×6×6 the corners are far away and the opening is about the centre. Up to symmetry Black again has two first moves: a face-adjacent move (the Satellite Opening, the engine's choice) and a diagonal one (the Comet Opening). The book has 70 analysed positions. The deepest named line is 7 plies long (Orion — Hunter), and from its last position the book still supplies the next reply.

Satellite Opening

`B-3-V` (1 move; discs 6–3; White to move)

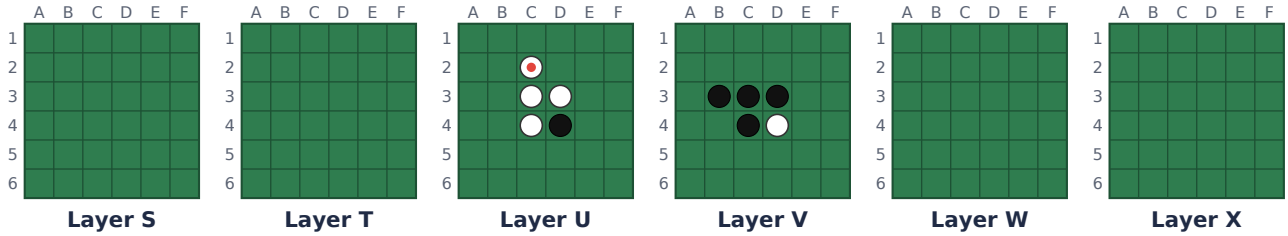


Black plays on a face next to the centre block. On 6×6×6 this is the engine's choice; the corners are too far away to matter yet.

Best replies (White): `C-2-U`, `E-4-U`, `D-4-T`

Orion

`B-3-V C-2-U` (2 moves; discs 5–5; Black to move)

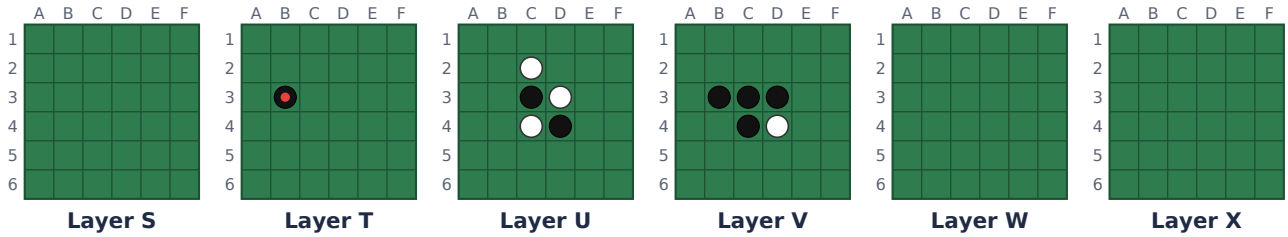


White's best reply: it sits diagonally beside Black's new disc and keeps the position balanced.

Best replies (Black): `B-3-T`, `D-2-U`, `C-2-T`

Orion — Belt

`B-3-V C-2-U B-3-T` (3 moves; discs 7–4; White to move)

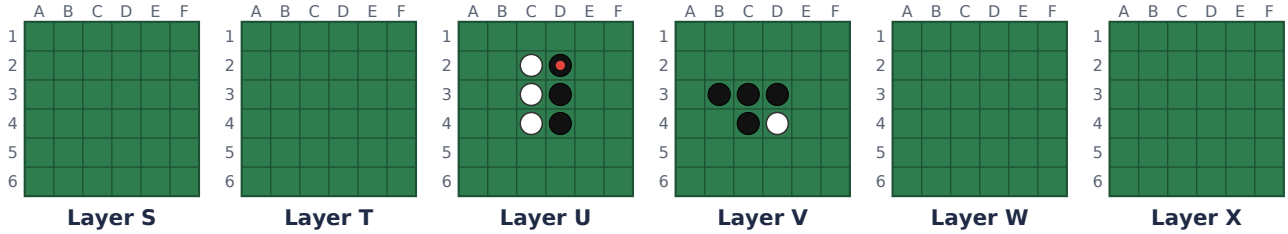


Black's main answer: B-3-T mirrors Black's first disc two layers lower, like the stars of Orion's belt. White continues E-4-U or D-4-T.

Best replies (White): `E-4-U`, `D-4-T`, `D-5-U`

Orion — Sword

B-3-V C-2-U D-2-U (3 moves; discs 7–4; White to move)

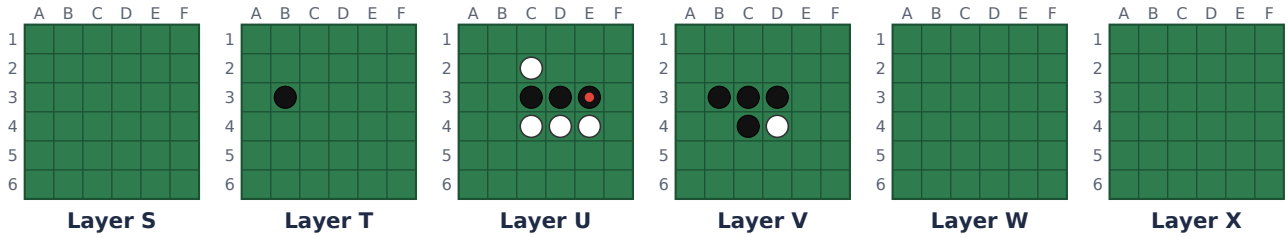


An equal alternative for Black; White's best is E-3-U.

Best replies (White): E-3-U, B-4-V, E-1-U

Orion — Nebula

B-3-V C-2-U B-3-T E-4-U E-3-U (5 moves; discs 8–5; White to move)

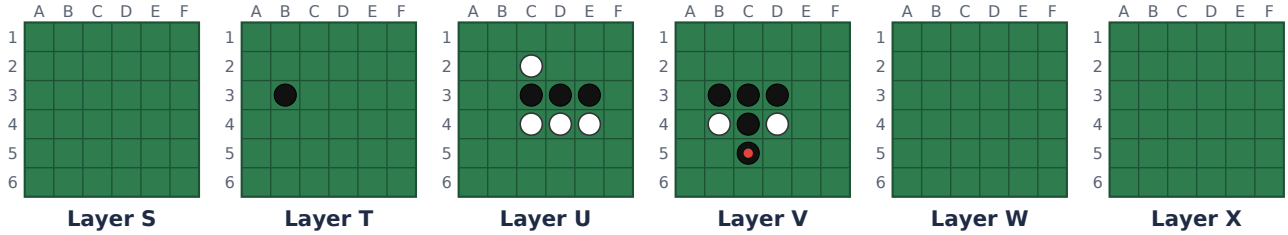


The main line after five moves: both sides spread through the middle layers. White's best is B-4-V.

Best replies (White): B-4-V, D-2-U, D-2-T

Orion — Hunter

B-3-V C-2-U B-3-T E-4-U E-3-U B-4-V C-5-V (7 moves; discs 9–6; White to move)

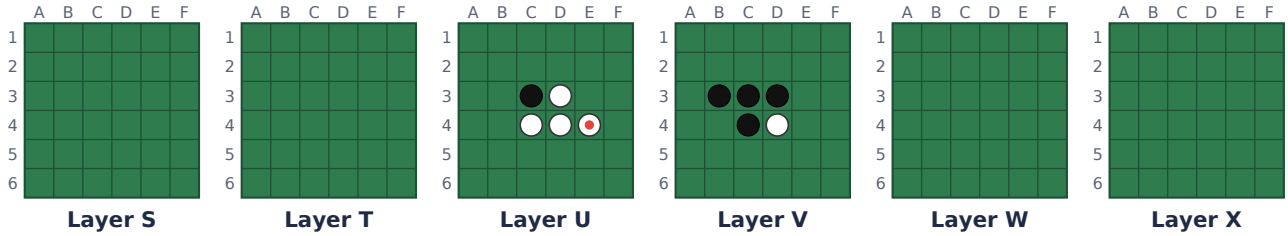


The deepest 6×6×6 main line (7 plies). White continues B-4-W.

Best replies (White): B-4-W, D-2-U, B-2-U

Andromeda

B-3-V E-4-U (2 moves; discs 5–5; Black to move)

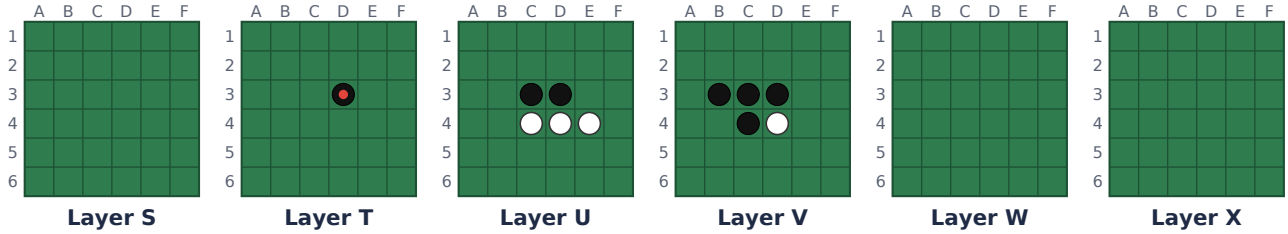


White answers on the far side of the centre block. Black replies D-3-T or E-3-U.

Best replies (Black): D-3-T, E-3-U, E-3-T

Andromeda — Spiral

`B-3-V E-4-U D-3-T` (3 moves; discs 7–4; White to move)

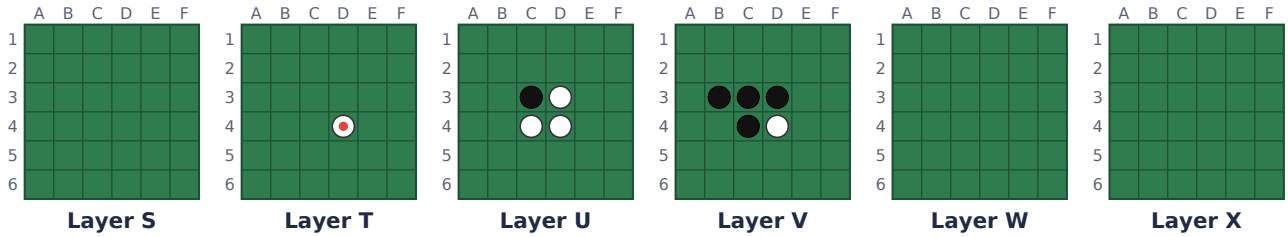


Black's best; White continues B-4-V or B-4-W.

Best replies (White): `B-4-V`, `B-4-W`, `D-2-U`

Pyramid

`B-3-V D-4-T` (2 moves; discs 5–5; Black to move)

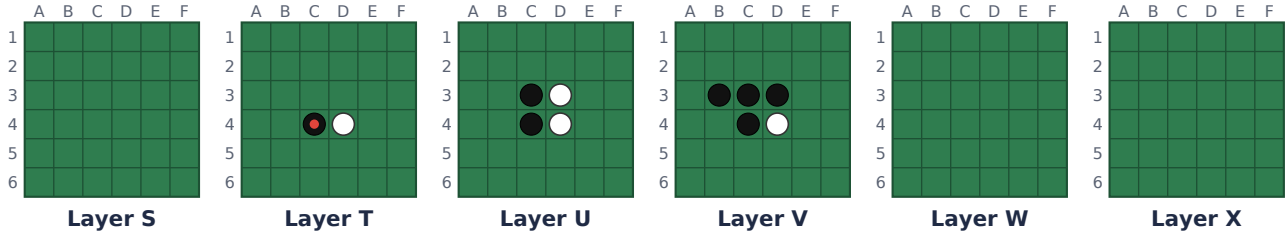


White builds upward through layer T; Black answers C-4-T.

Best replies (Black): `C-4-T`, `D-3-T`, `C-5-U`

Pyramid — Apex

`B-3-V D-4-T C-4-T` (3 moves; discs 7–4; White to move)

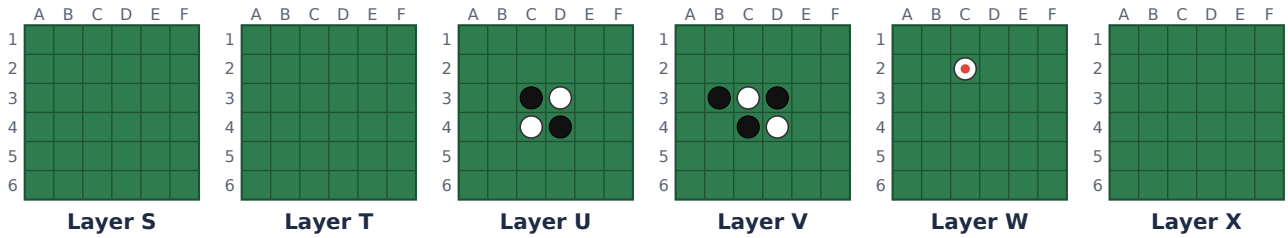


Black's best reply; the position stays dead level.

Best replies (White): `B-2-V`, `D-3-W`, `B-4-U`

Prism

`B-3-V C-2-W` (2 moves; discs 5–5; Black to move)

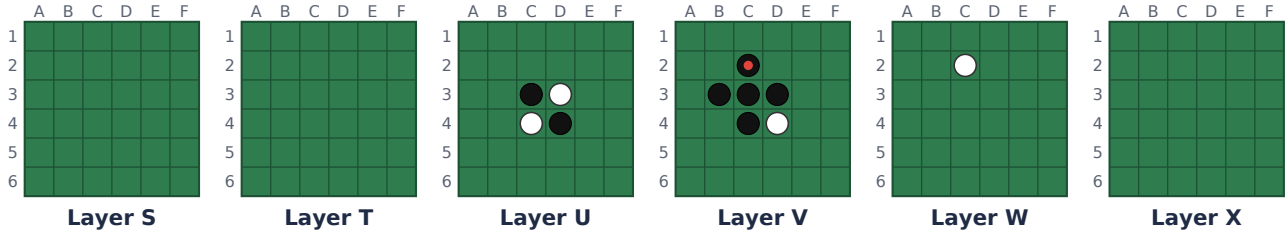


White plays into layer W; Black answers C-2-V or B-4-U.

Best replies (Black): `C-2-V`, `B-4-U`, `E-3-U`

Prism — Refraction

`B-3-V C-2-W C-2-V` (3 moves; discs 7–4; White to move)

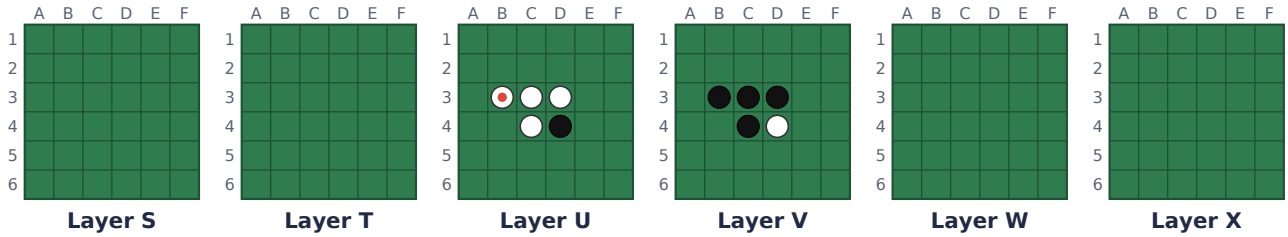


Black's best; White continues D-5-U.

Best replies (White): `D-5-U`, `D-3-W`, `B-3-W`

Domino

`B-3-V B-3-U` (2 moves; discs 5–5; Black to move)

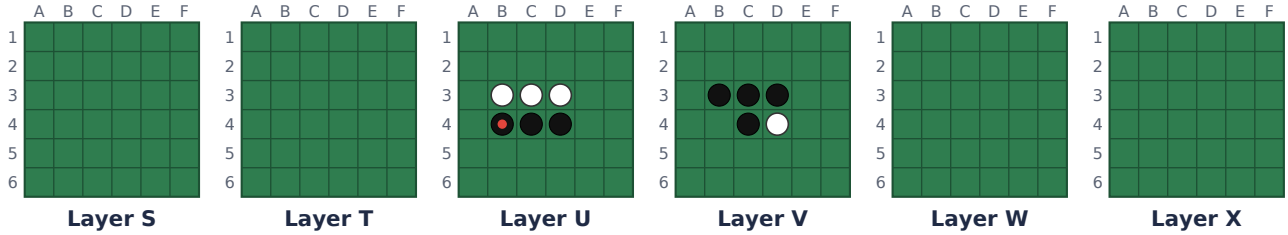


White plays right beside Black's first disc; Black answers B-4-U or B-2-U.

Best replies (Black): `B-4-U`, `B-2-U`, `D-4-W`

Domino — Double Six

`B-3-V B-3-U B-4-U` (3 moves; discs 7–4; White to move)

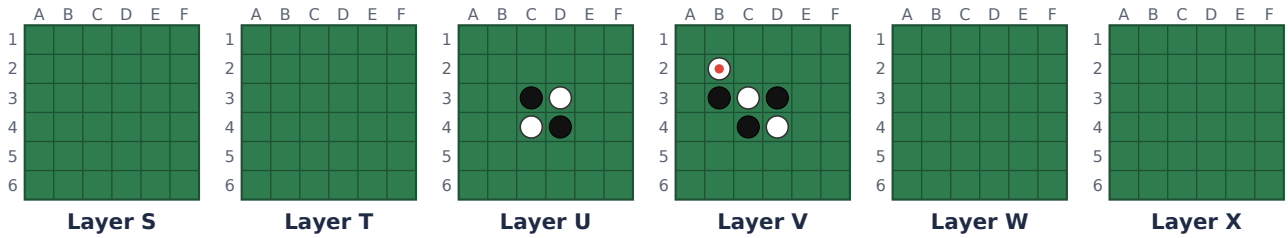


Black's best reply, extending the domino line. White continues B-4-V.

Best replies (White): `B-4-V`, `C-5-W`, `D-5-W`

Tetrahedron

`B-3-V B-2-V` (2 moves; discs 5–5; Black to move)

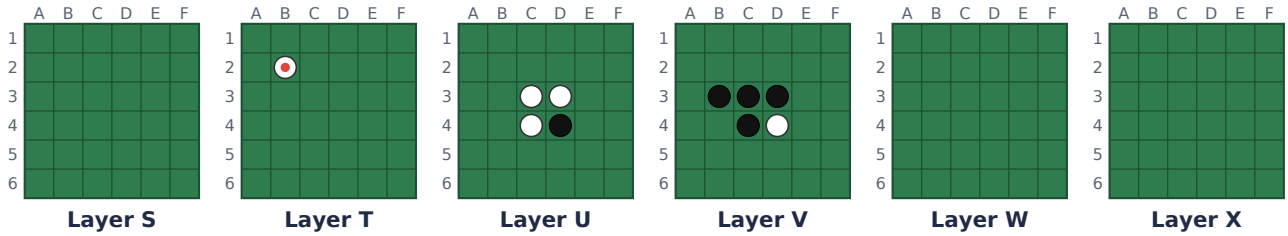


White also plays in column B; Black answers C-2-V or D-2-U.

Best replies (Black): `C-2-V`, `D-2-U`, `C-3-W`

Meteor Strike

`B-3-V B-2-T` (2 moves; discs 5–5; Black to move)

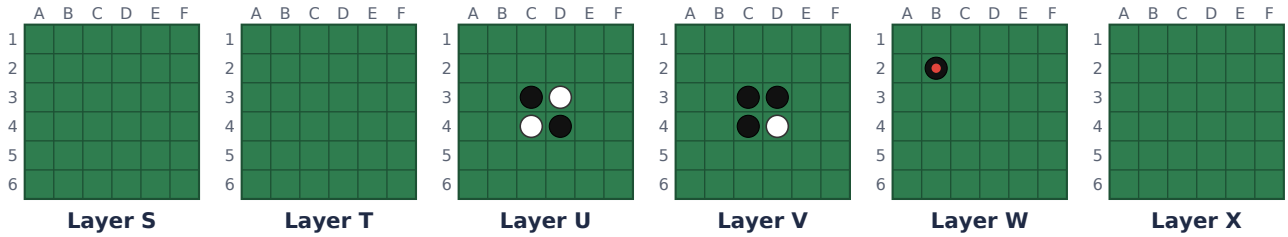


A poor reply that hands Black the initiative: the evaluation swings sharply to Black.

Best replies (Black): `C-3-T`, `C-2-T`, `B-3-T`

Comet Opening

`B-2-W` (1 move; discs 6–3; White to move)

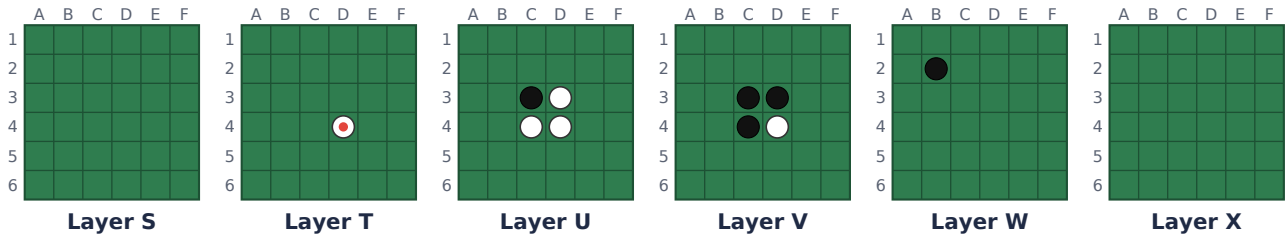


Black plays diagonally off the centre block. Playable, but the engine rates it clearly below the Satellite Opening.

Best replies (White): `D-4-T`, `B-2-V`, `B-3-U`

Halley

`B-2-W D-4-T` (2 moves; discs 5–5; Black to move)

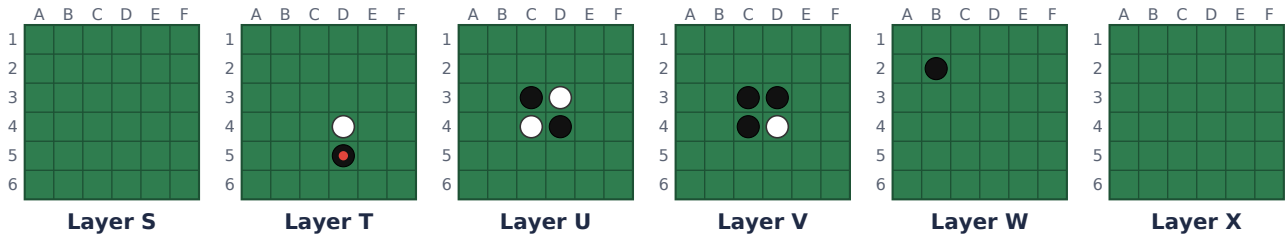


White's best answer to the Comet Opening. Black replies D-5-T.

Best replies (Black): `D-5-T`

Halley — Perihelion

`B-2-W D-4-T D-5-T` (3 moves; discs 7–4; White to move)

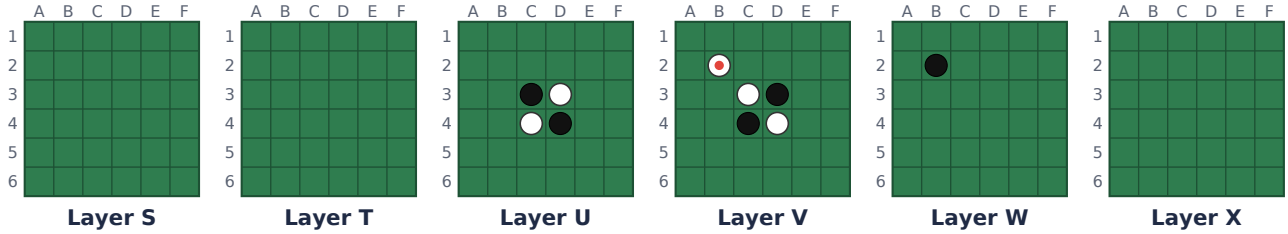


Black's best reply; White continues C-4-W or C-2-U.

Best replies (White): `C-4-W`, `C-2-U`, `D-5-U`

Shooting Star

`B-2-W B-2-V` (2 moves; discs 5–5; Black to move)

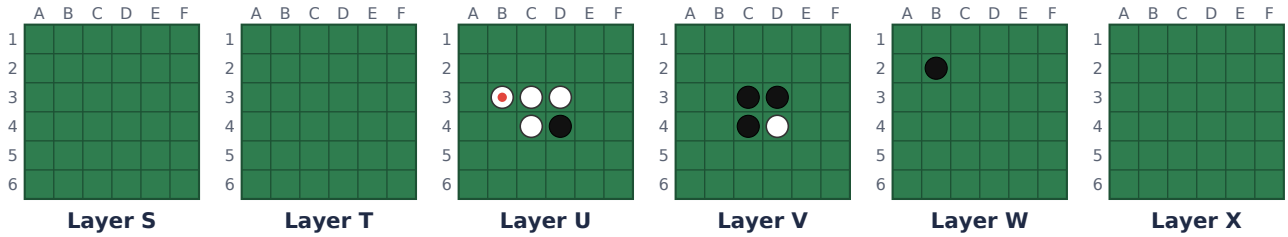


White also plays in column B; nearly as good as Halley. Black answers C-3-W.

Best replies (Black): `C-3-W`, `B-3-V`

Stardust

`B-2-W B-3-U` (2 moves; discs 5–5; Black to move)

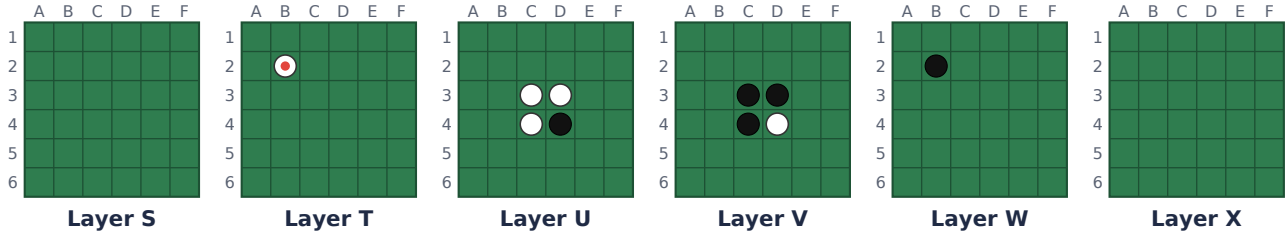


White plays B-3-U, next to the centre block; Black answers D-4-W or B-4-U.

Best replies (Black): `D-4-W`, `B-4-U`, `C-4-T`

Supernova

`B-2-W B-2-T` (2 moves; discs 5–5; Black to move)



A weak reply: Black answers E-5-V with a large advantage.

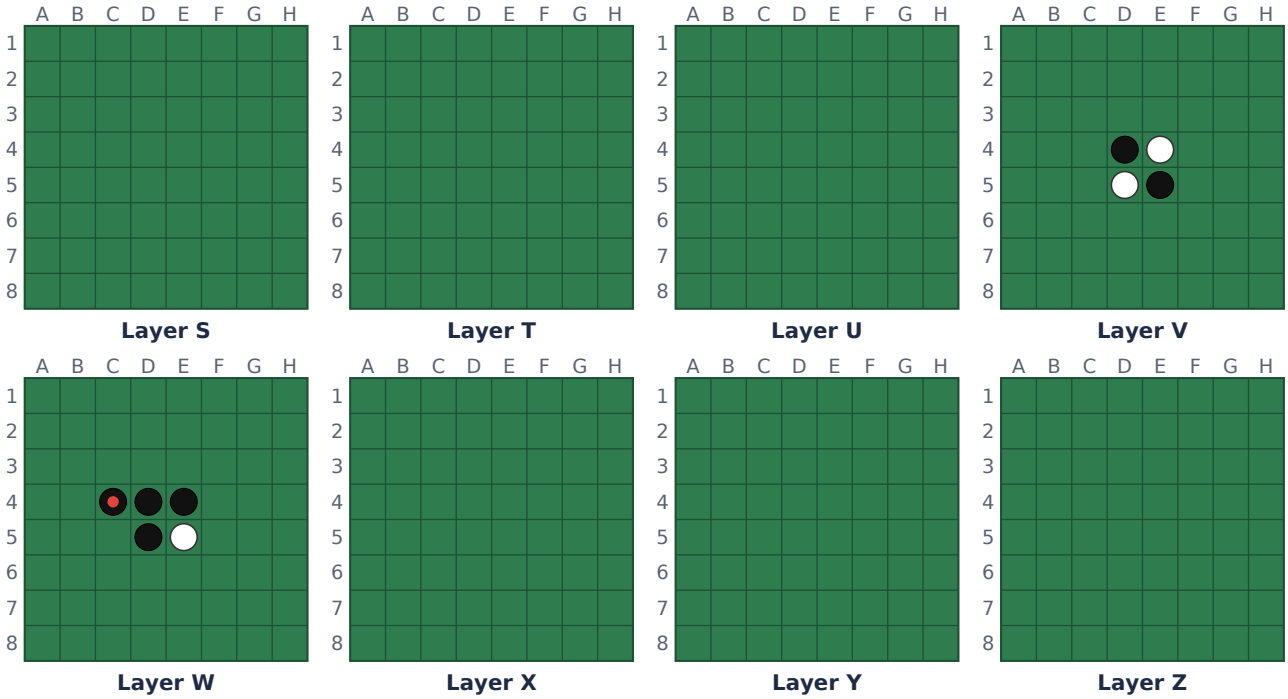
Best replies (Black): `E-5-V`, `D-4-W`

Part 4 — 8×8×8 cube openings

On the largest cube the opening is slow and positional; the engine rates almost all early moves within a few points of each other, so the book is shallower and mainly saves the Bot thinking time. Black's two first moves up to symmetry are the face-adjacent Quadrant Opening and the diagonal Lattice Opening. The book has 66 analysed positions. The deepest named line is 7 plies long (Skyscraper — Penthouse), and from its last position the book still supplies the next reply.

Quadrant Opening

`C-4-W` (1 move; discs 6–3; White to move)

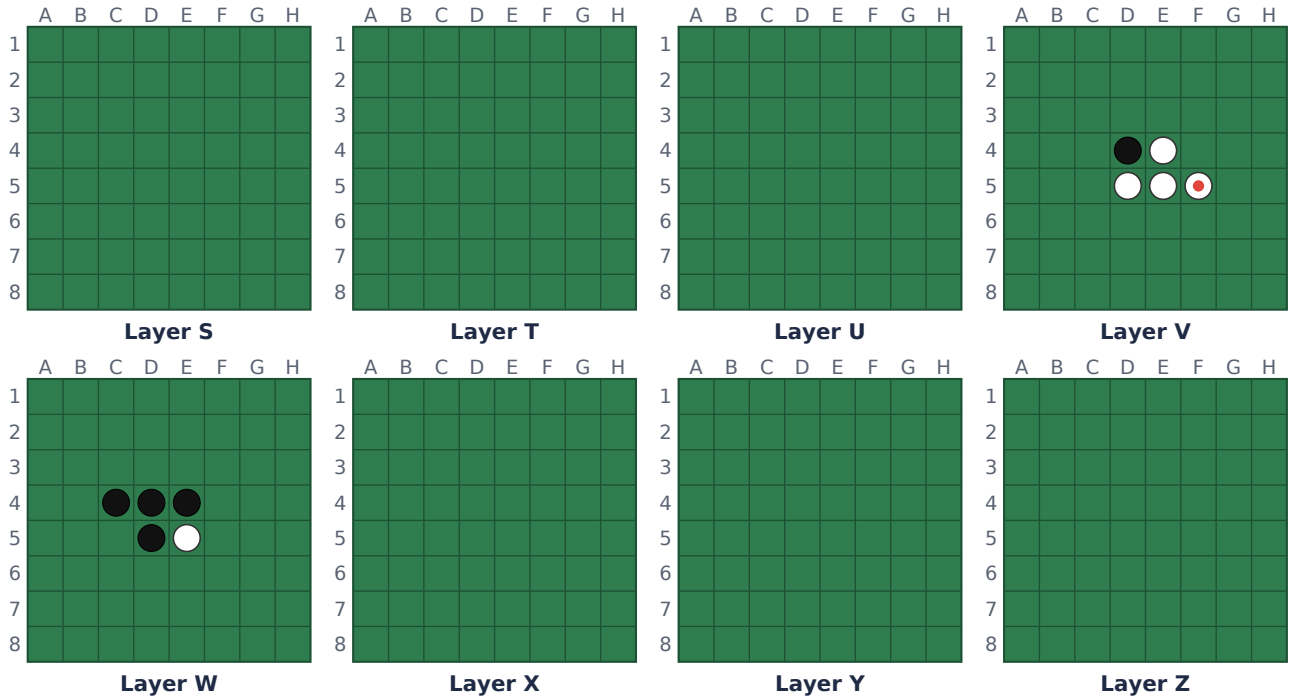


Black plays on a face next to the centre block. The engine's first choice on 8×8×8, by a whisker.

Best replies (White): `F-5-V`, `D-3-V`, `C-4-V`

Skyscraper

`C-4-W F-5-V` (2 moves; discs 5–5; Black to move)

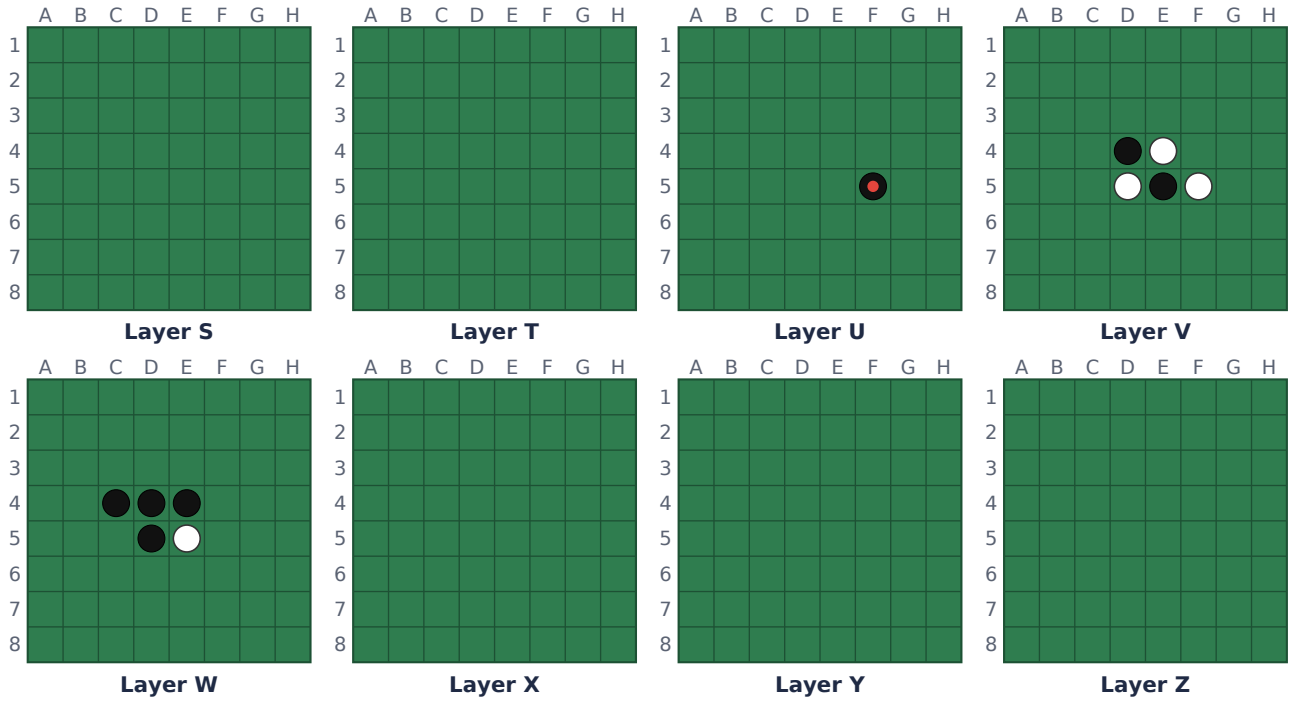


One of several equal White replies; play builds across the middle layers. Black answers F-5-U or F-3-U.

Best replies (Black): F-5-U, F-3-U, F-4-V

Skyscraper — Elevator

`C-4-W F-5-V F-5-U` (3 moves; discs 7–4; White to move)

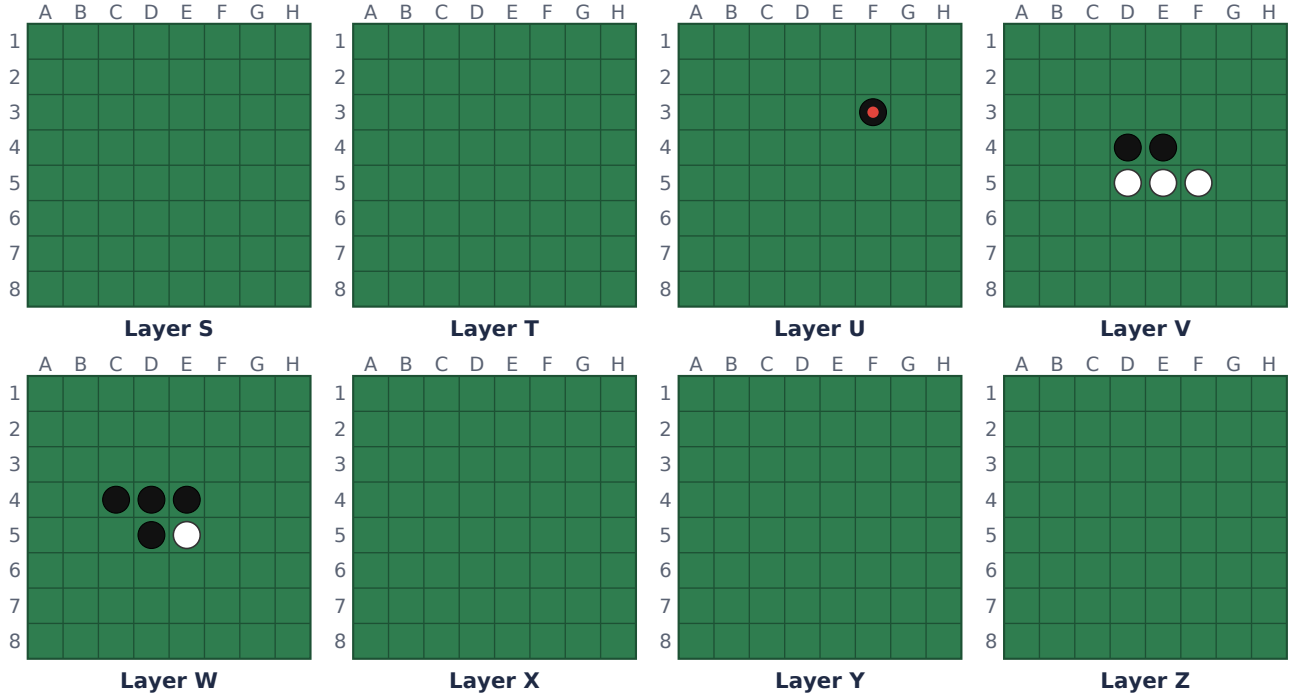


Black's main reply, stacking straight below White's disc. White continues E-3-W or E-4-X.

Best replies (White): E-3-W, E-4-X, E-5-U

Skyscraper — Fire Escape

`C-4-W F-5-V F-3-U` (3 moves; discs 7–4; White to move)

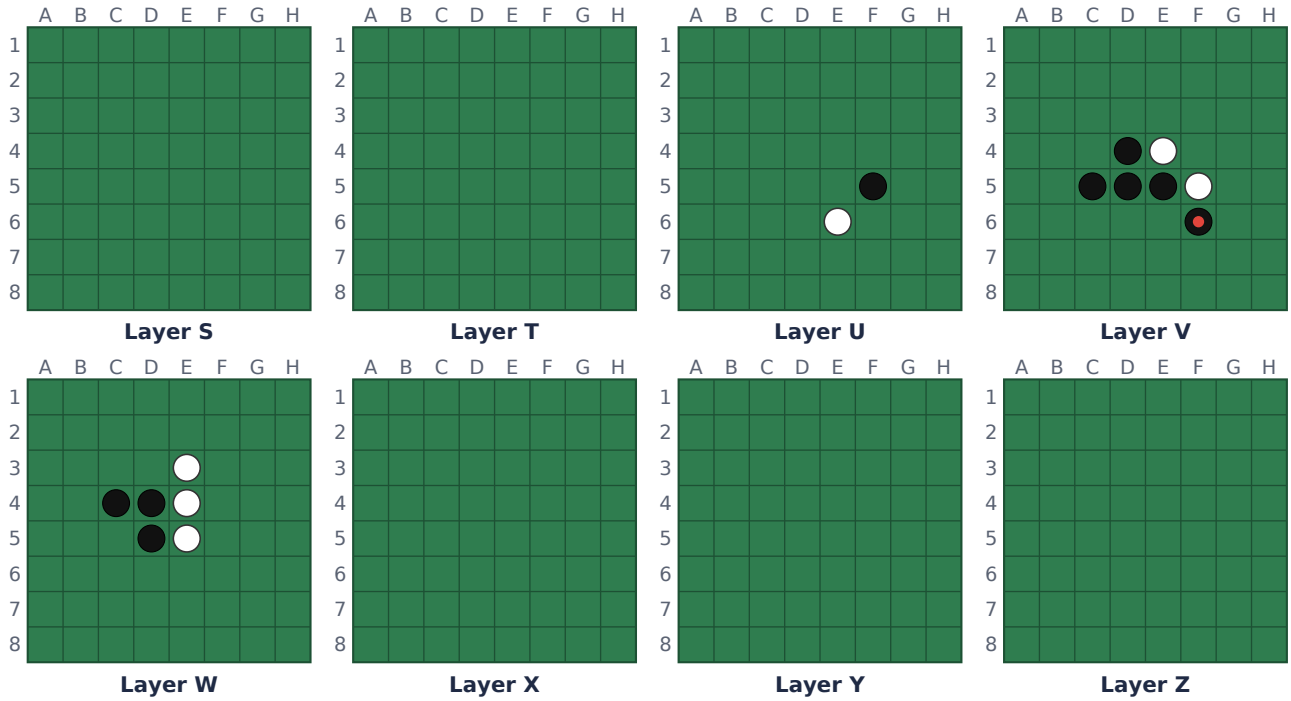


An equal alternative for Black; White continues E-3-W.

Best replies (White): `E-3-W`, `E-3-X`, `E-3-V`

Skyscraper — Penthouse

`C-4-W F-5-V F-5-U E-3-W C-5-V E-6-U F-6-V` (7 moves; discs 9–6; White to move)

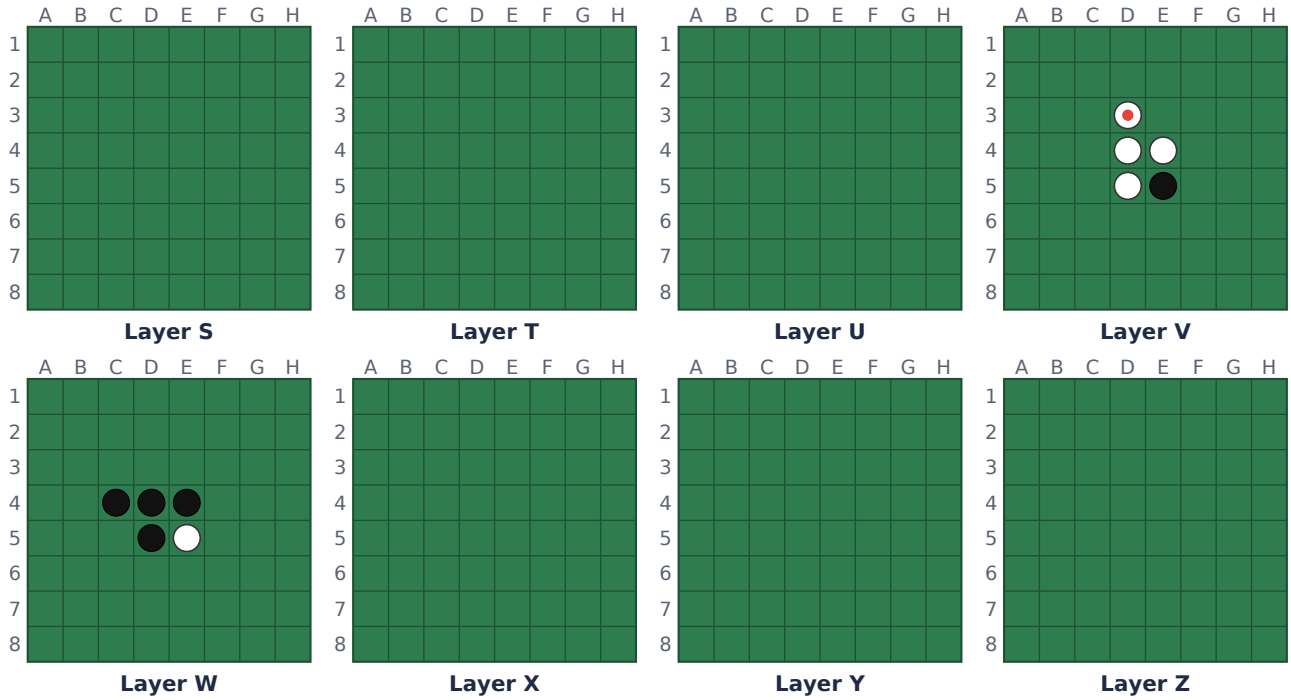


The deepest 8×8×8 main line (7 plies). White continues E-6-V.

Best replies (White): `E-6-V`, `C-5-W`, `C-4-X`

Cathedral

`C-4-W D-3-V` (2 moves; discs 5–5; Black to move)

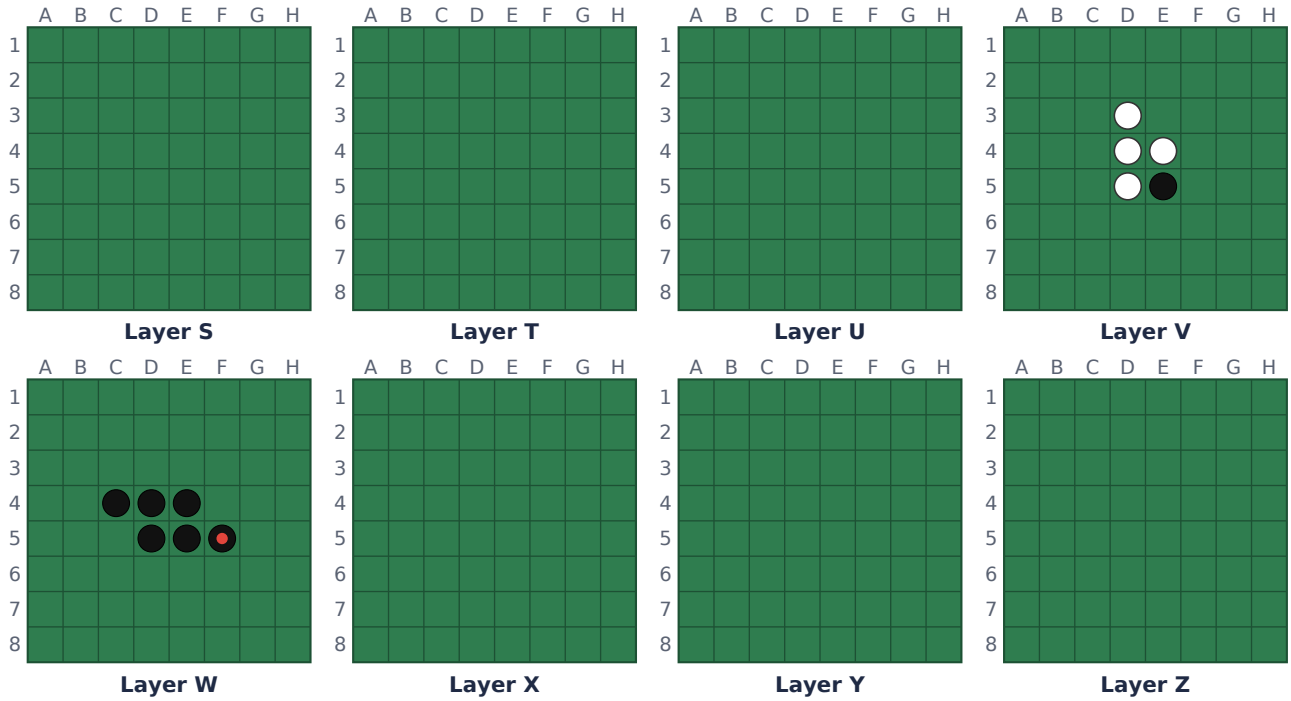


An equal White reply that lands in the centre layers. Black answers F-5-W or F-6-W.

Best replies (Black): `F-5-W`, `F-6-W`, `E-5-X`

Cathedral — Nave

`C-4-W D-3-V F-5-W` (3 moves; discs 7–4; White to move)

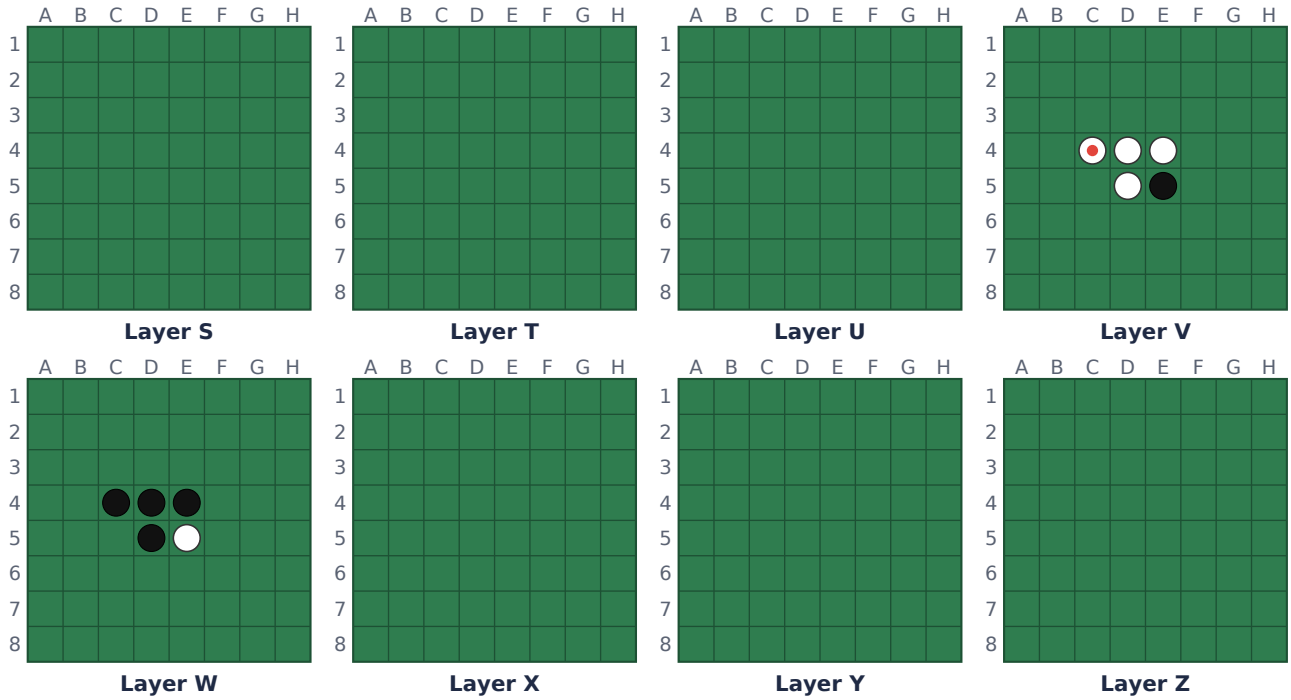


Black's best; White continues F-6-V.

Best replies (White): `F-6-V`, `E-6-V`, `D-4-X`

Tower

`C-4-W C-4-V` (2 moves; discs 5–5; Black to move)

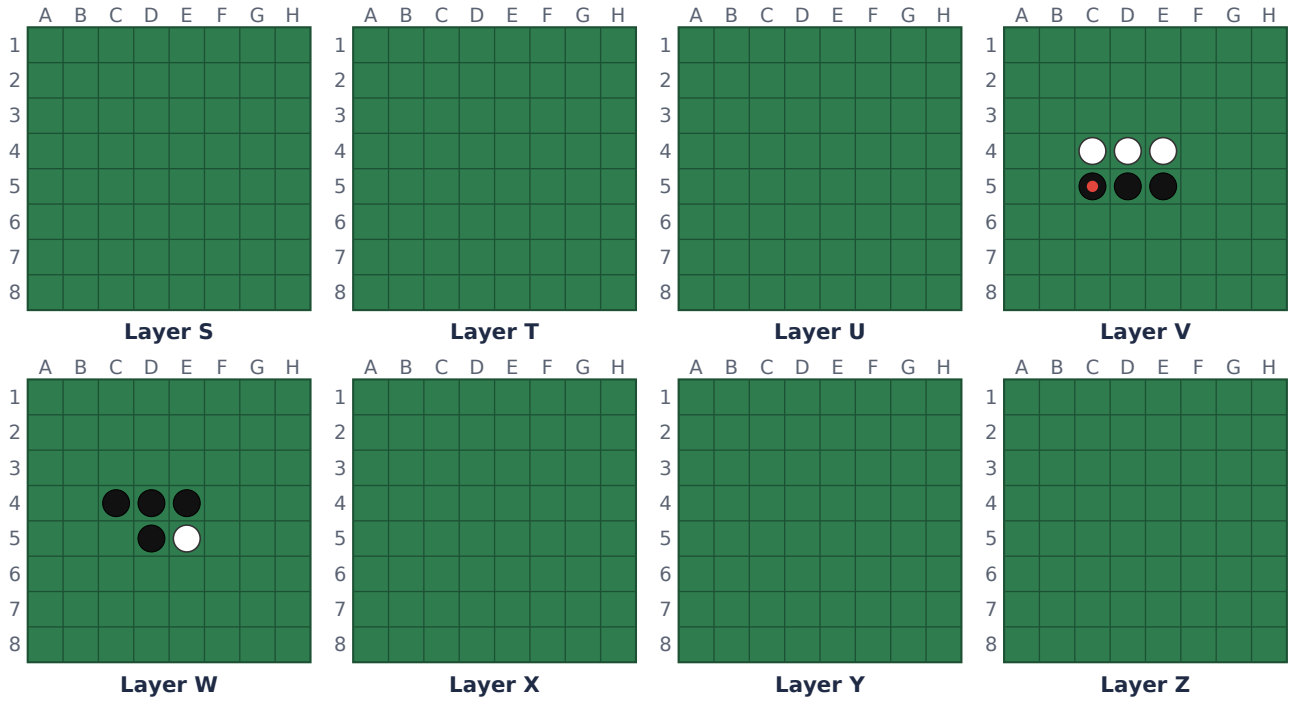


White plays straight next to Black's first disc. Black answers C-5-V or D-4-U.

Best replies (Black): `C-5-V`, `D-4-U`, `E-5-X`

Tower — Spire

`C-4-W C-4-V C-5-V` (3 moves; discs 7–4; White to move)

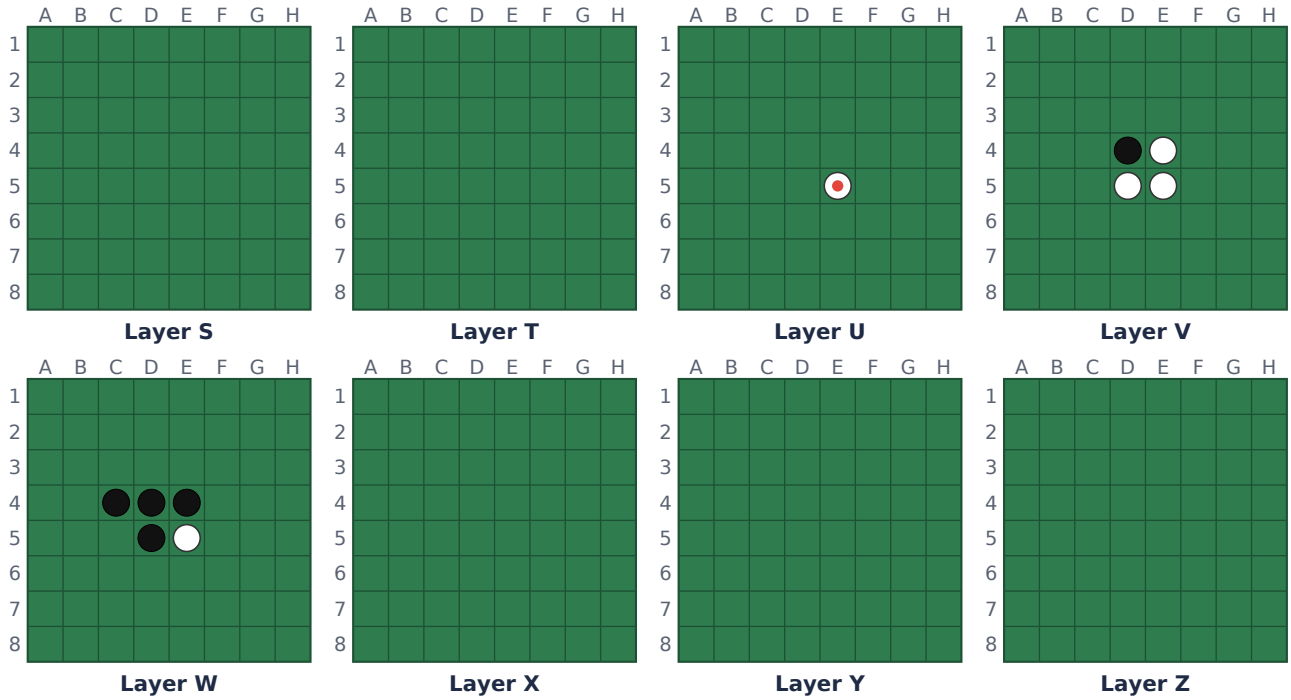


Black's best; White continues D-4-X.

Best replies (White): `D-4-X`, `C-5-W`, `C-3-W`

Pagoda

`C-4-W E-5-U` (2 moves; discs 5–5; Black to move)

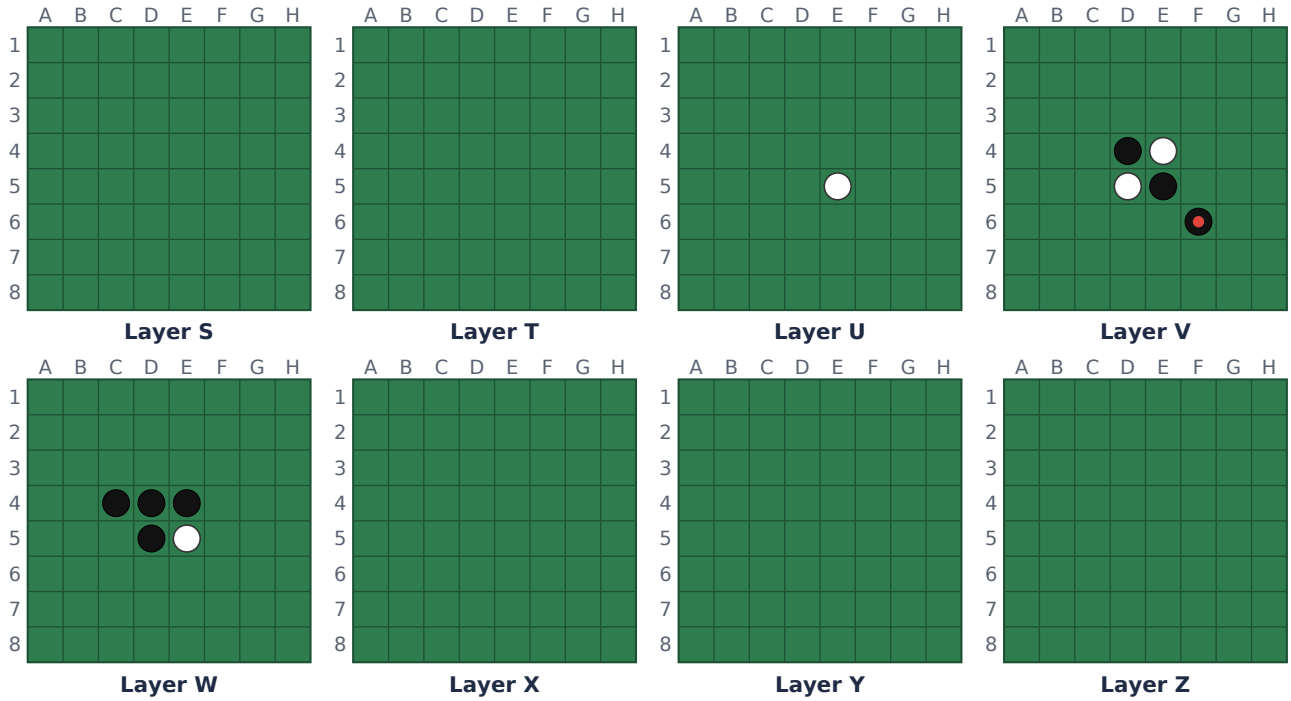


White answers from below the centre block. Black's best is F-6-V.

Best replies (Black): `F-6-V`, `F-4-V`, `D-6-V`

Pagoda — Roof

`C-4-W E-5-U F-6-V` (3 moves; discs 7–4; White to move)

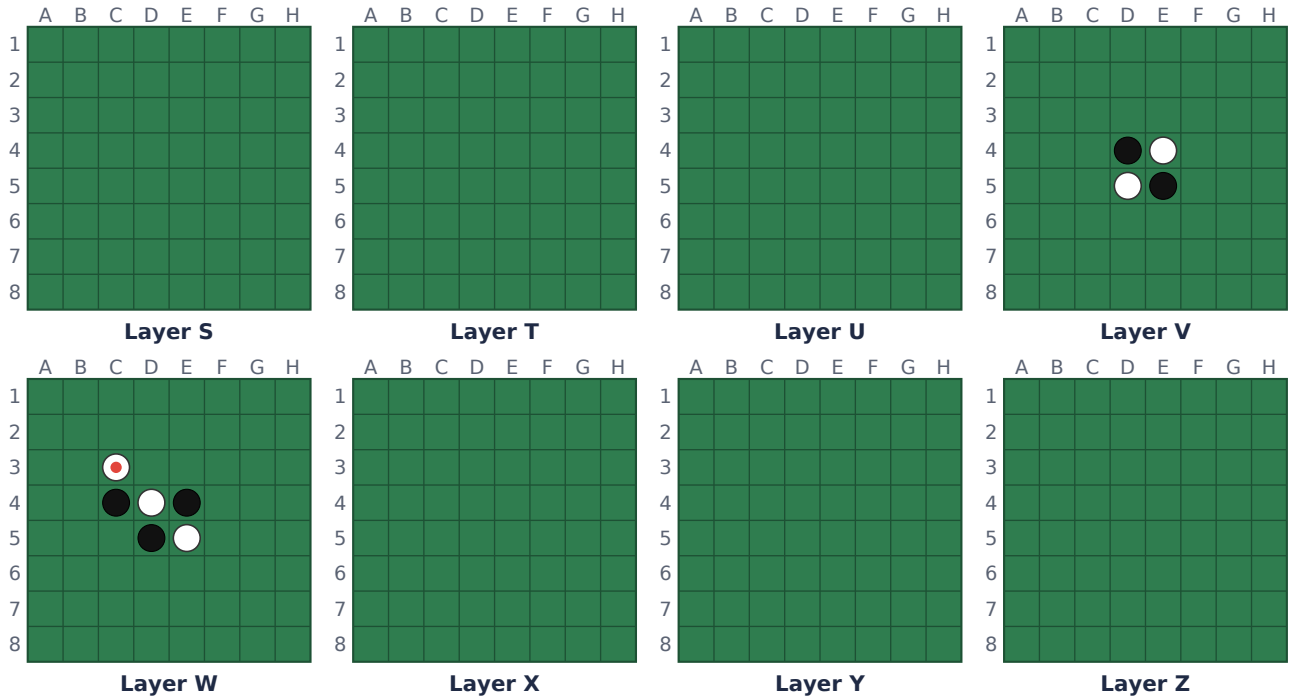


Black's main reply; White continues E-6-V.

Best replies (White): `E-6-V`, `F-5-V`, `D-5-X`

Labyrinth

`C-4-W C-3-W` (2 moves; discs 5–5; Black to move)

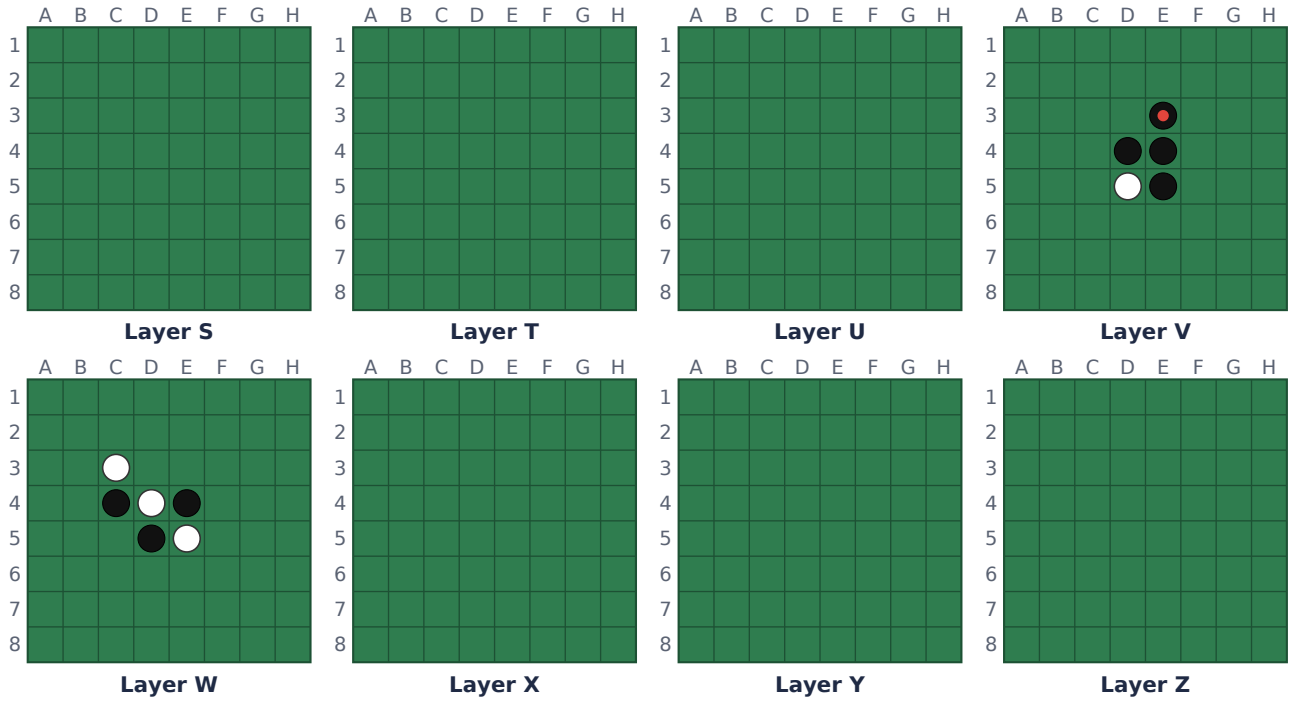


White plays beside Black in the same layer. Black answers E-3-V or D-4-X.

Best replies (Black): `E-3-V`, `D-4-X`, `D-3-W`

Labyrinth — Minotaur

`C-4-W C-3-W E-3-V` (3 moves; discs 7–4; White to move)

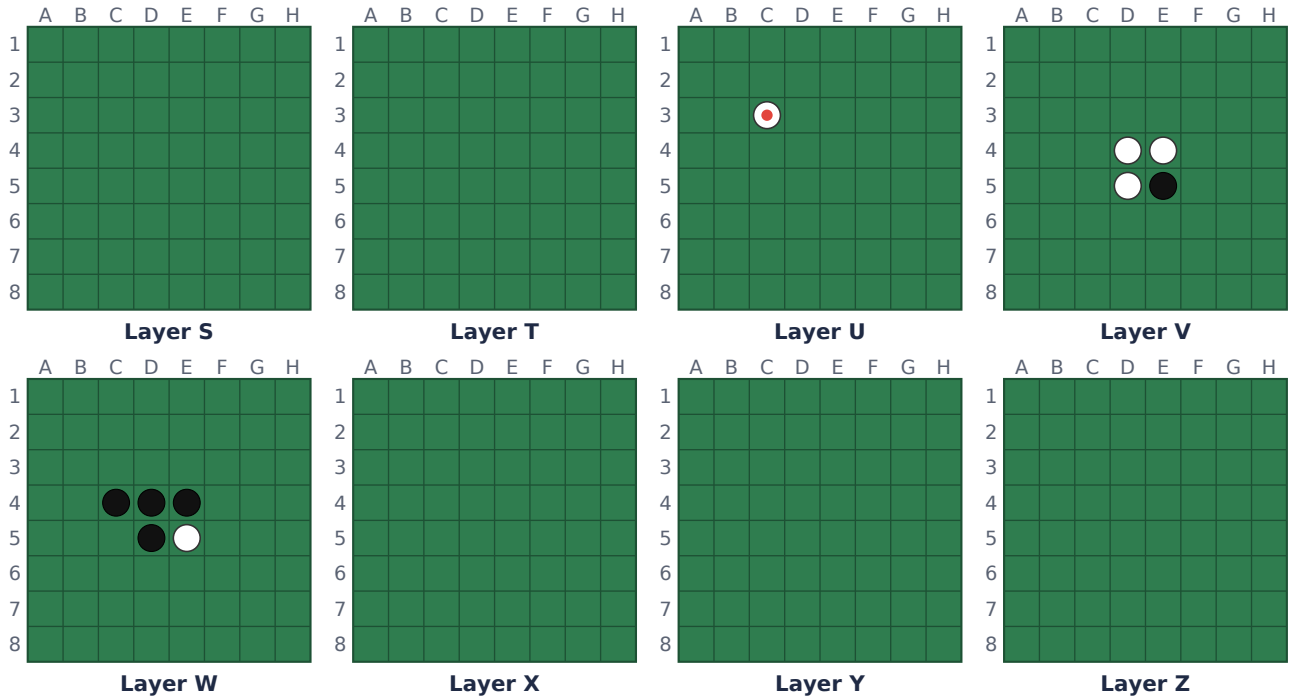


Black's best; White continues D-6-W.

Best replies (White): `D-6-W`, `F-4-W`, `D-4-U`

Bridge

`C-4-W C-3-U` (2 moves; discs 5–5; Black to move)

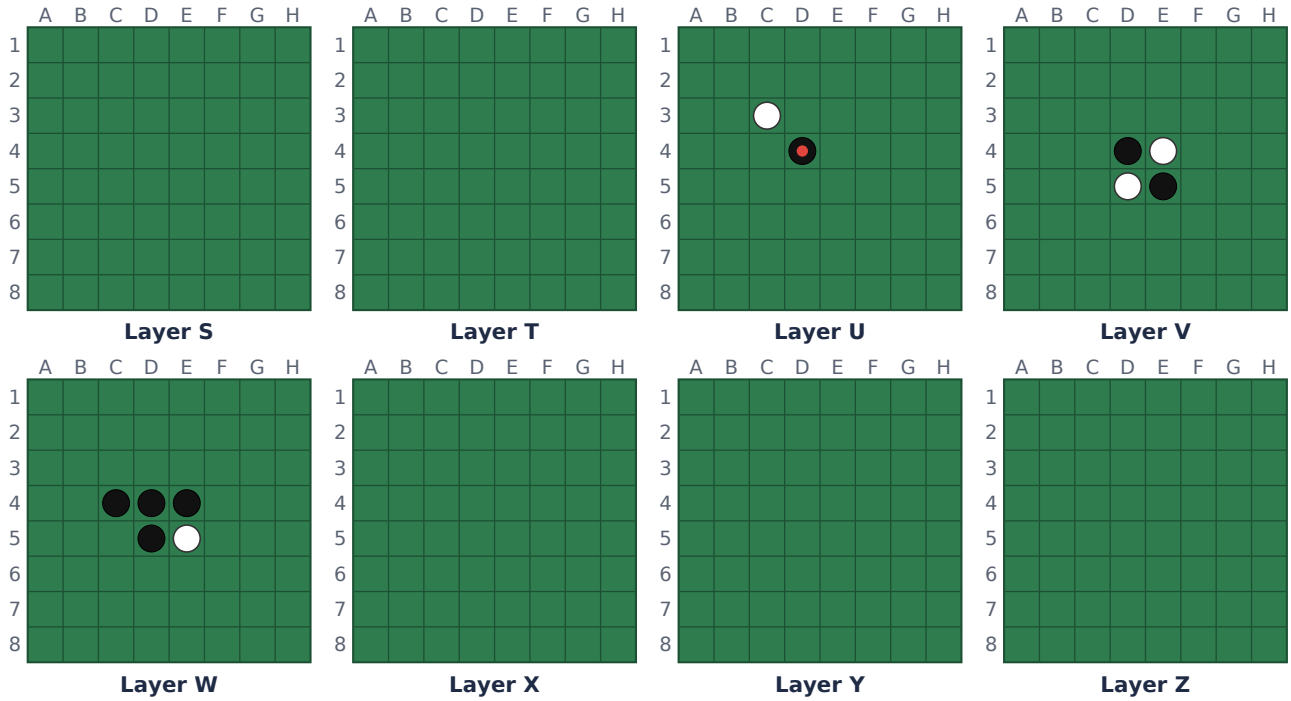


A sound White reply two layers away; Black answers D-4-U or E-4-U.

Best replies (Black): `D-4-U`, `E-4-U`, `C-3-V`

Bridge — Arch

`C-4-W C-3-U D-4-U` (3 moves; discs 7–4; White to move)

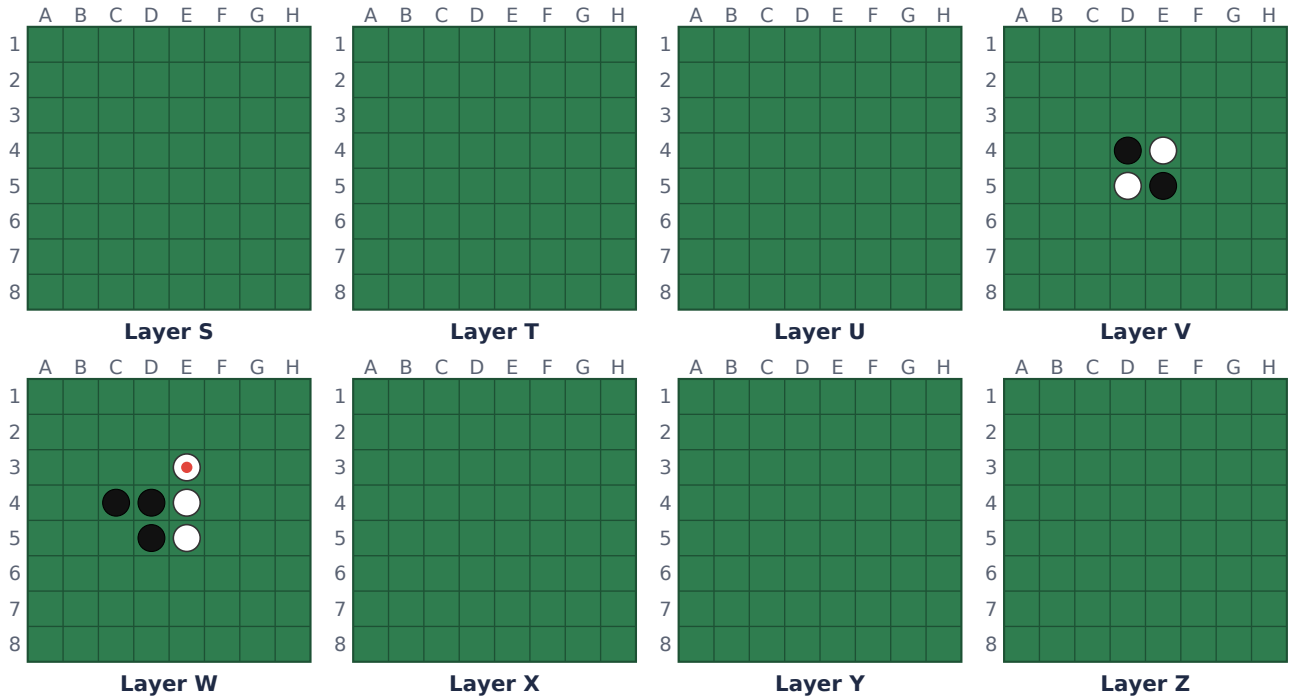


Black's best; White continues D-5-X.

Best replies (White): `D-5-X`, `F-5-V`, `C-4-V`

Colonnade

`C-4-W E-3-W` (2 moves; discs 5–5; Black to move)

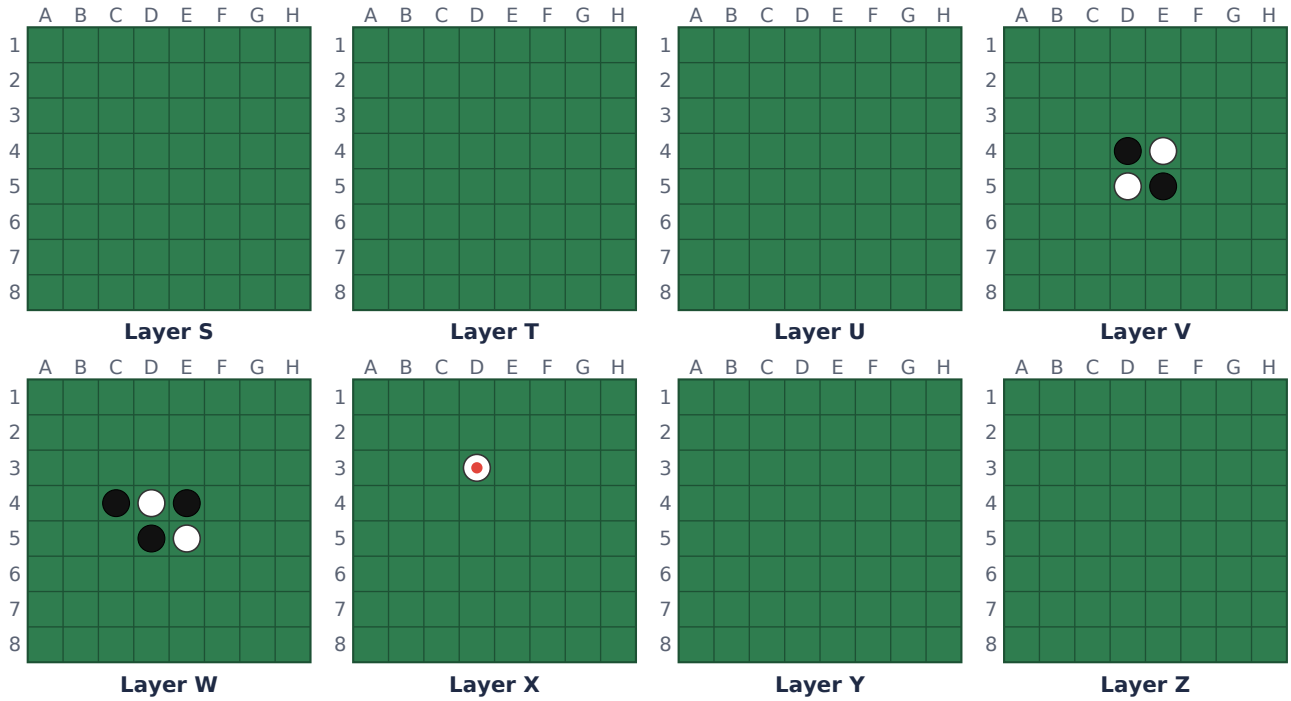


A sound White reply; Black answers D-6-V or D-5-U.

Best replies (Black): `D-6-V`, `D-5-U`, `C-5-V`

Ziggurat

`C-4-W D-3-X` (2 moves; discs 5–5; Black to move)

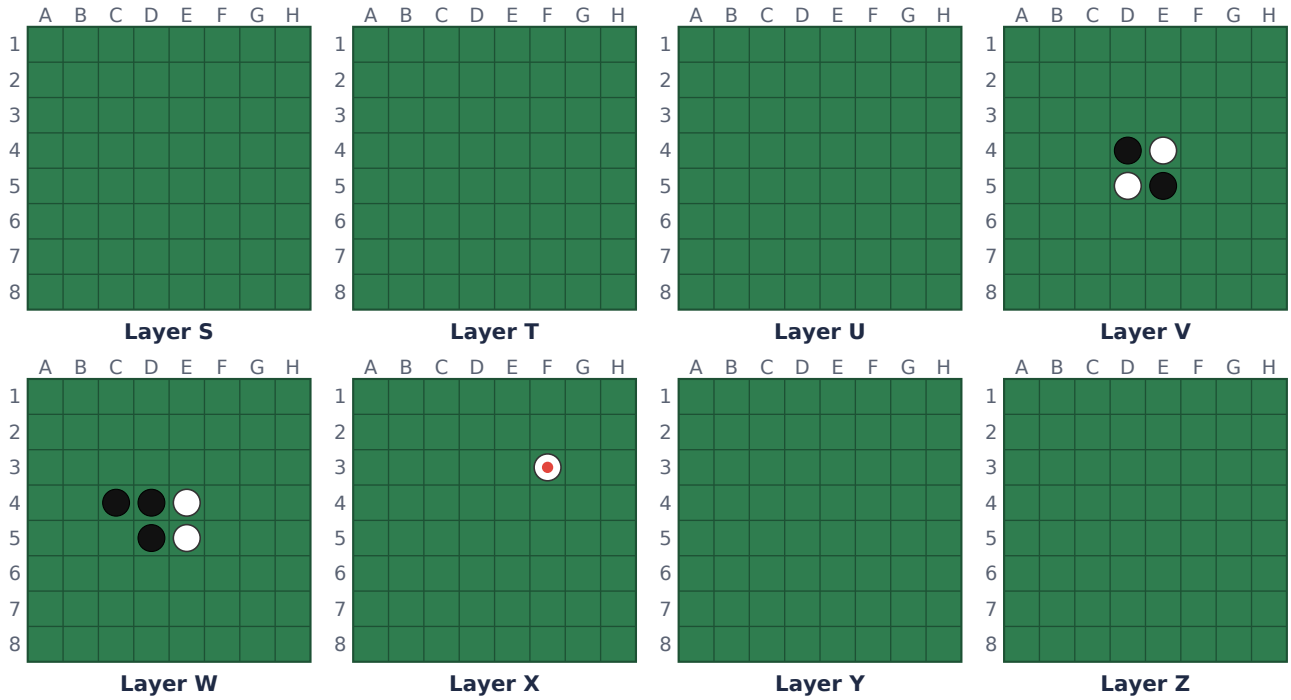


White reaches up into layer X. Black answers C-3-X or E-3-V with a slight edge.

Best replies (Black): `C-3-X`, `E-3-V`, `E-4-U`

Aqueduct

`C-4-W F-3-X` (2 moves; discs 5–5; Black to move)

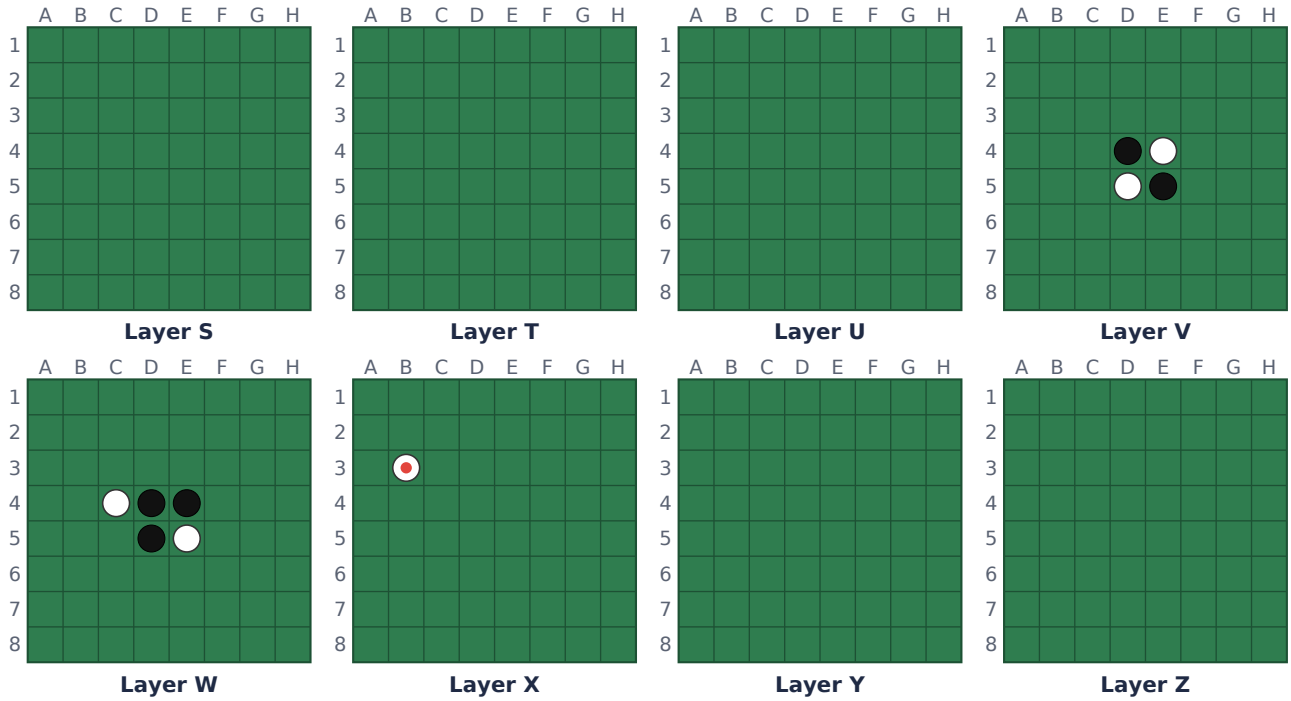


A weaker reply: Black answers C-5-V and is better.

Best replies (Black): `C-5-V`, `E-6-U`, `D-5-U`

Folly

`C-4-W B-3-X` (2 moves; discs 5–5; Black to move)

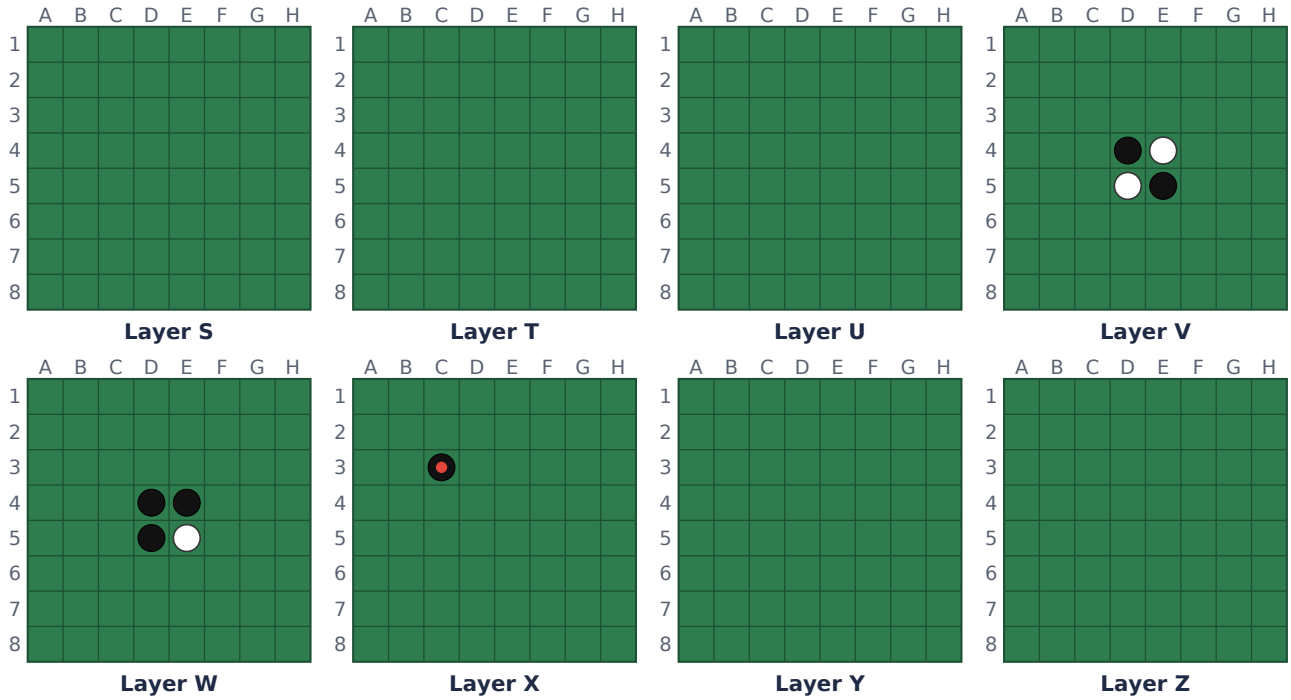


A weaker reply: Black answers E-3-V or C-5-V and is better.

Best replies (Black): `E-3-V`, `C-5-V`, `E-4-U`

Lattice Opening

`C-3-X` (1 move; discs 6–3; White to move)

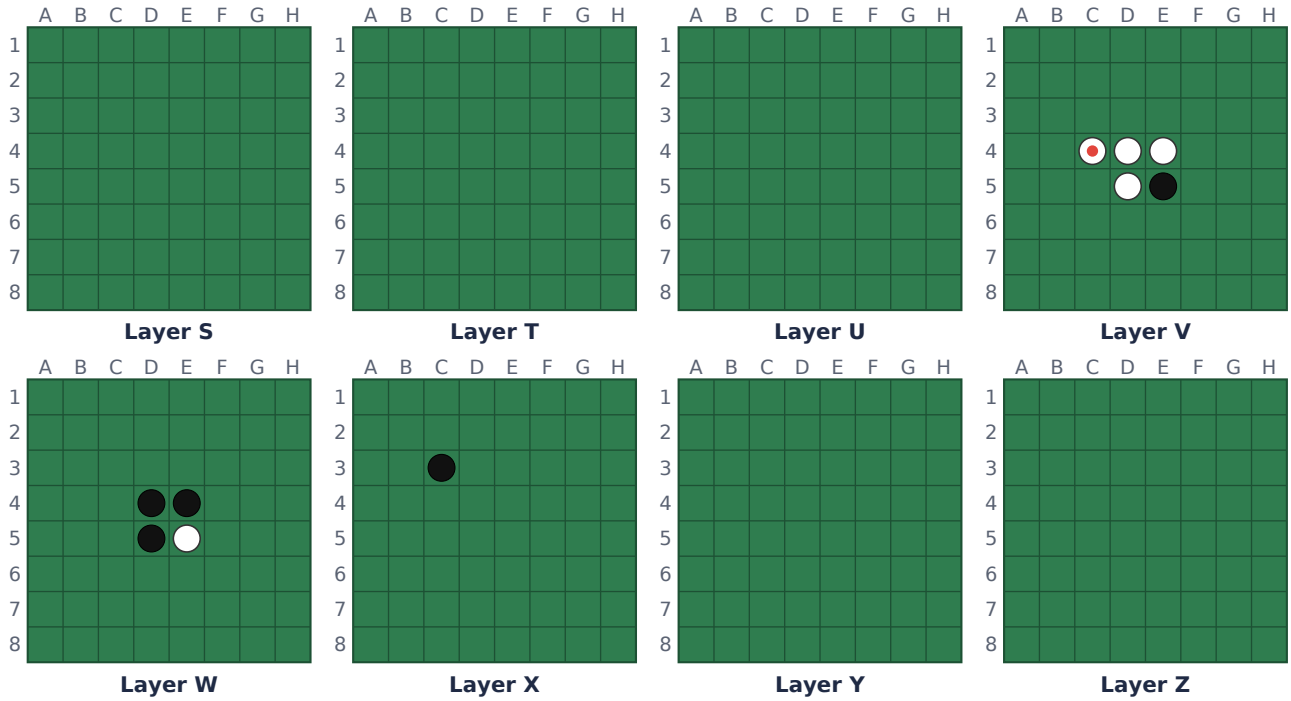


Black plays diagonally off a corner of the centre block. Practically equal to the Quadrant Opening.

Best replies (White): `C-4-V`, `E-5-U`, `C-3-U`

Crystal

`C-3-X C-4-V` (2 moves; discs 5–5; Black to move)

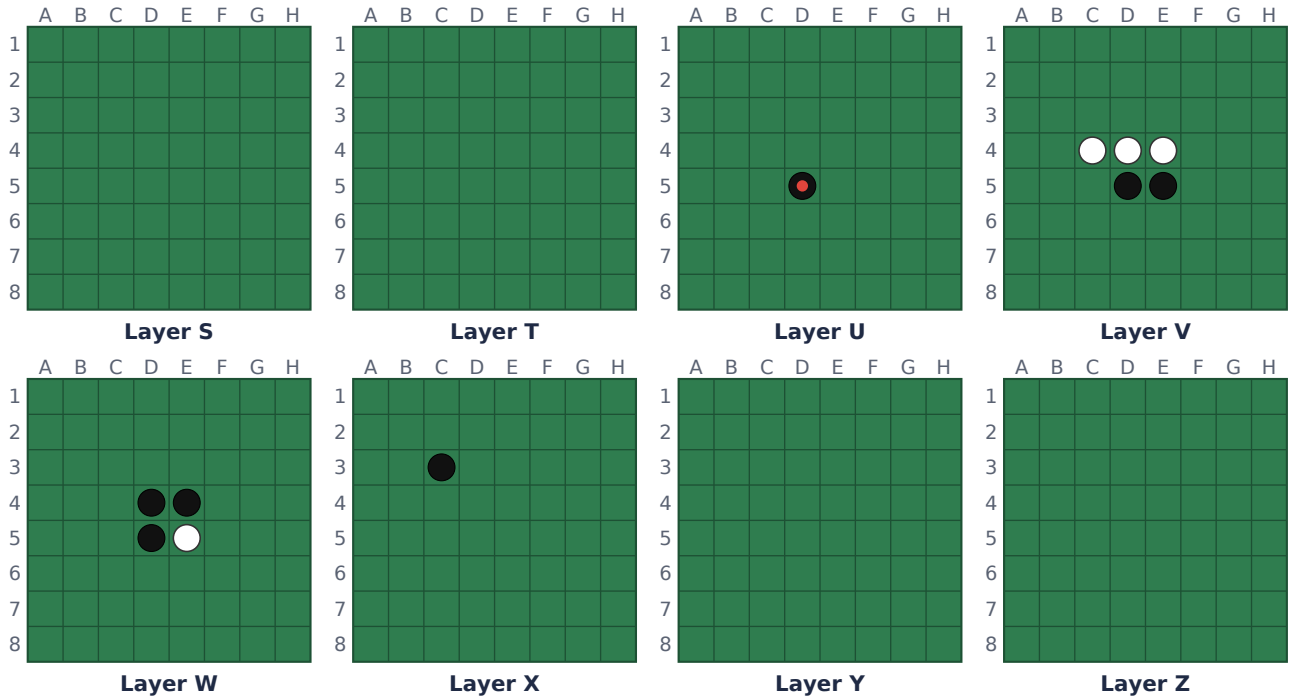


White's best answer to the Lattice Opening. Black replies D-5-U.

Best replies (Black): `F-5-W`, `D-5-U`, `D-4-U`

Crystal — Facet

`C-3-X C-4-V D-5-U` (3 moves; discs 7–4; White to move)

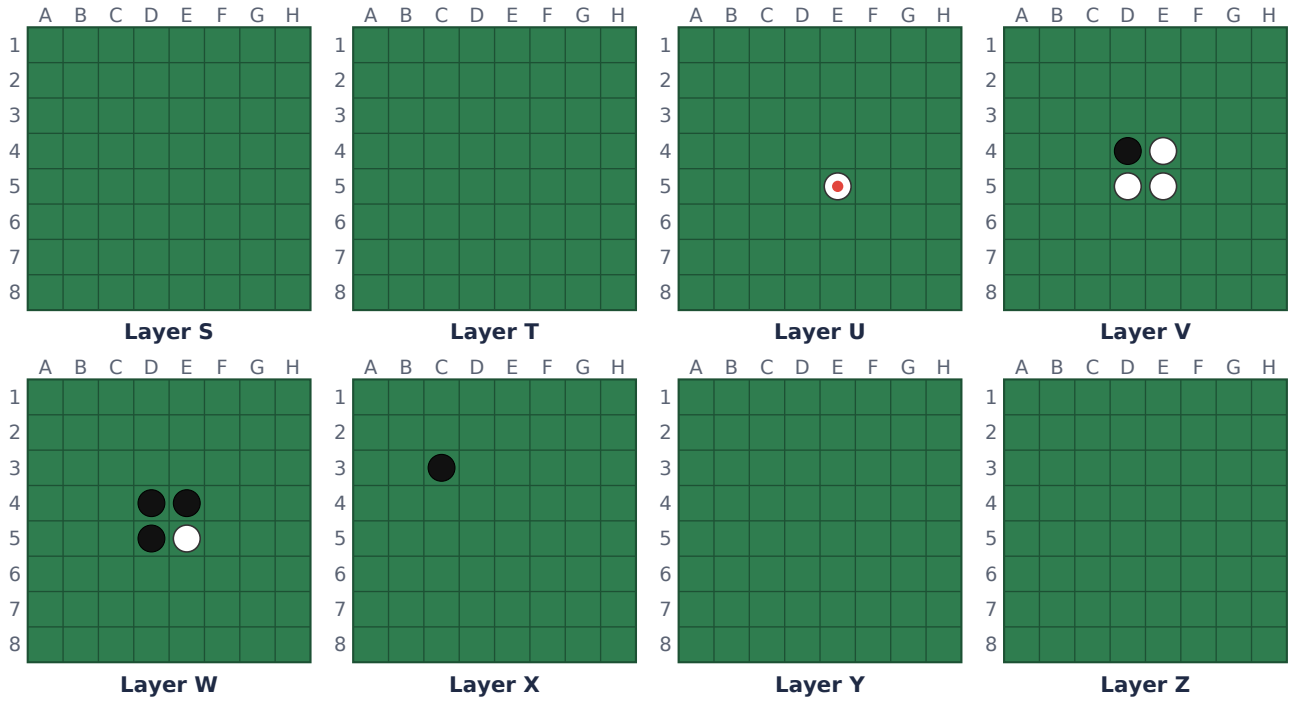


Black's main reply; White continues E-3-W or F-4-X.

Best replies (White): `E-3-W`, `F-4-X`, `D-4-X`

Diamond

`C-3-X E-5-U` (2 moves; discs 5–5; Black to move)

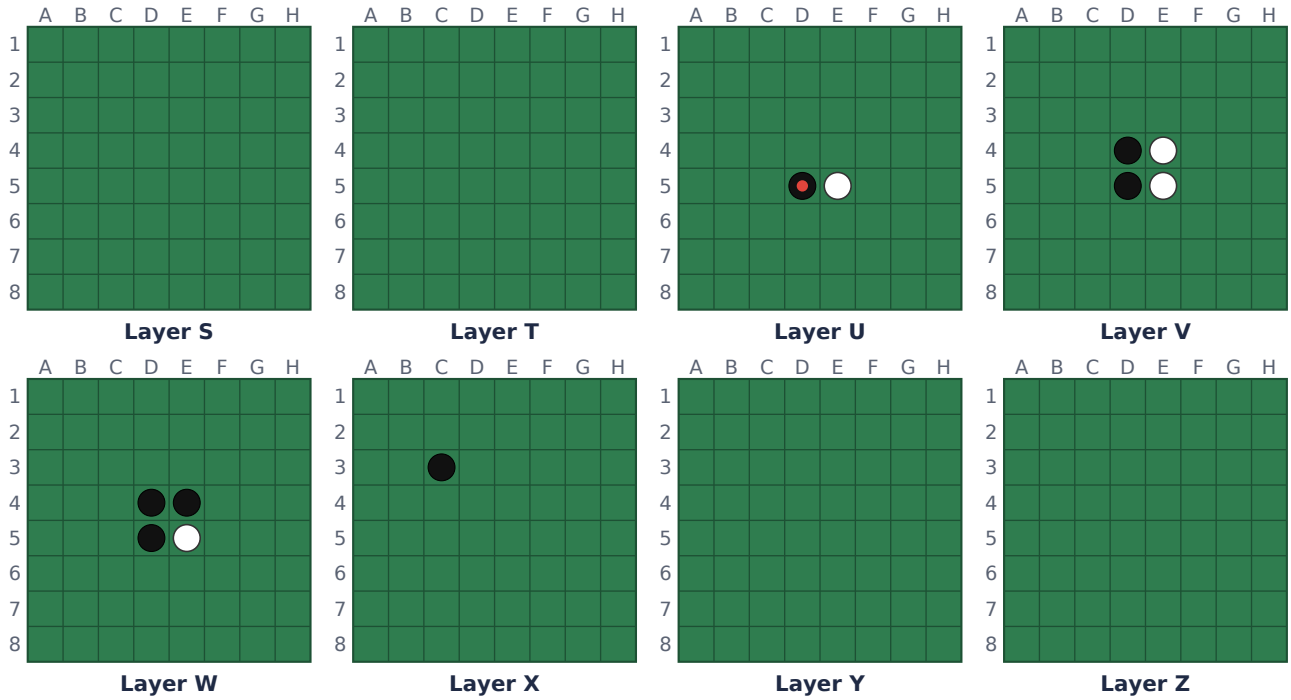


A good White alternative; Black replies D-5-U.

Best replies (Black): `D-5-U`, `D-6-V`, `F-6-V`

Diamond — Cut

`C-3-X E-5-U D-5-U` (3 moves; discs 7–4; White to move)

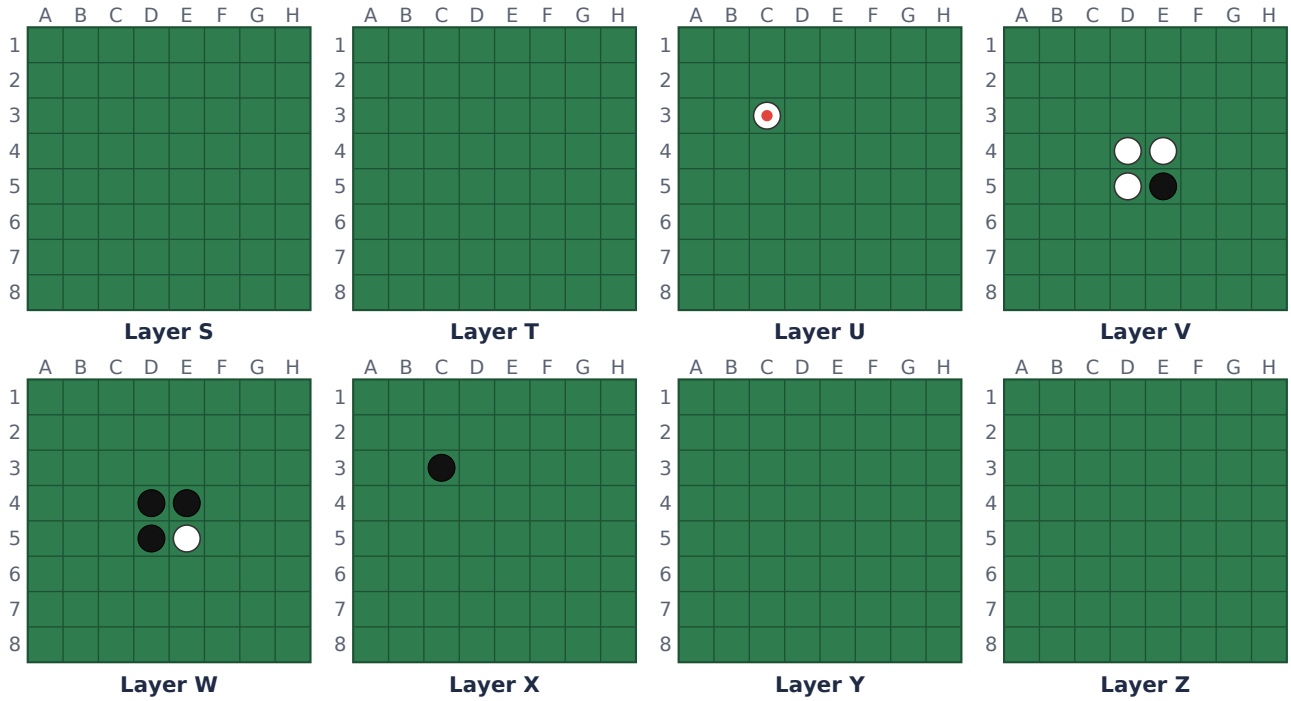


Black's best; White continues C-5-V.

Best replies (White): `C-5-V`, `E-3-X`, `C-4-V`

Graphite

`C-3-X C-3-U` (2 moves; discs 5–5; Black to move)

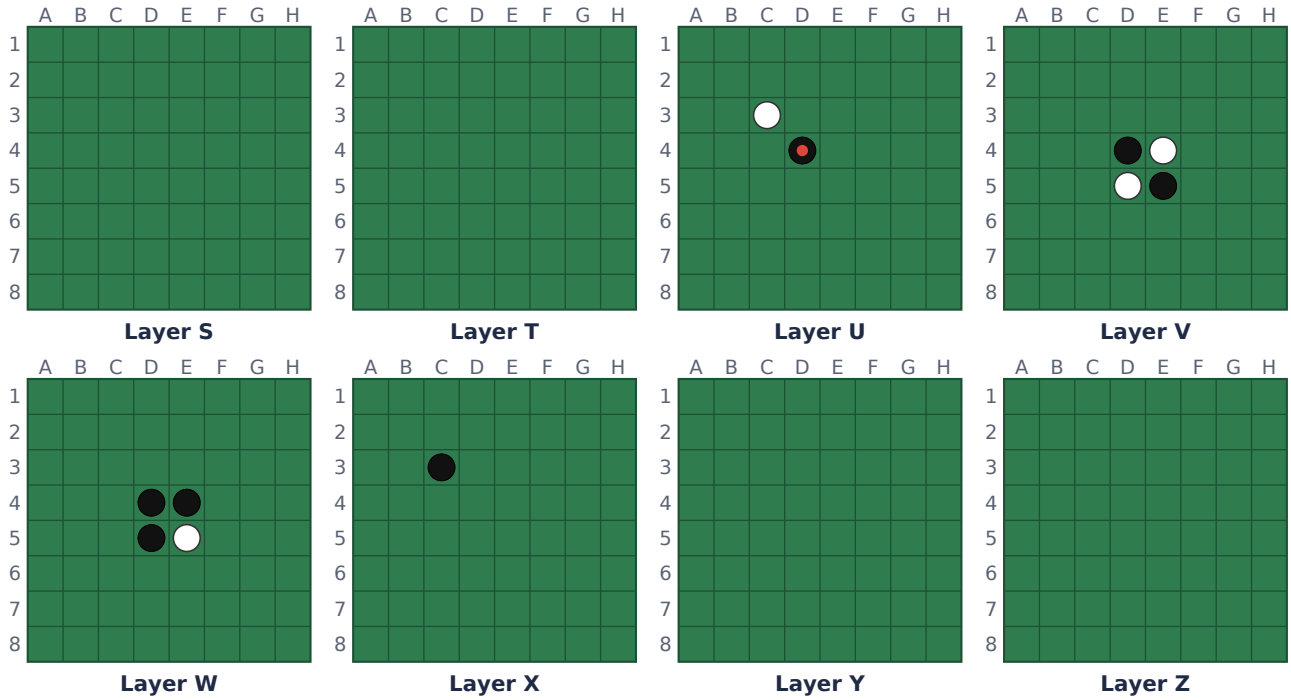


White plays in the same column, three layers lower. Black answers D-4-U.

Best replies (Black): `D-4-U`, `C-3-V`, `E-5-X`

Graphite — Layer

`C-3-X C-3-U D-4-U` (3 moves; discs 7–4; White to move)

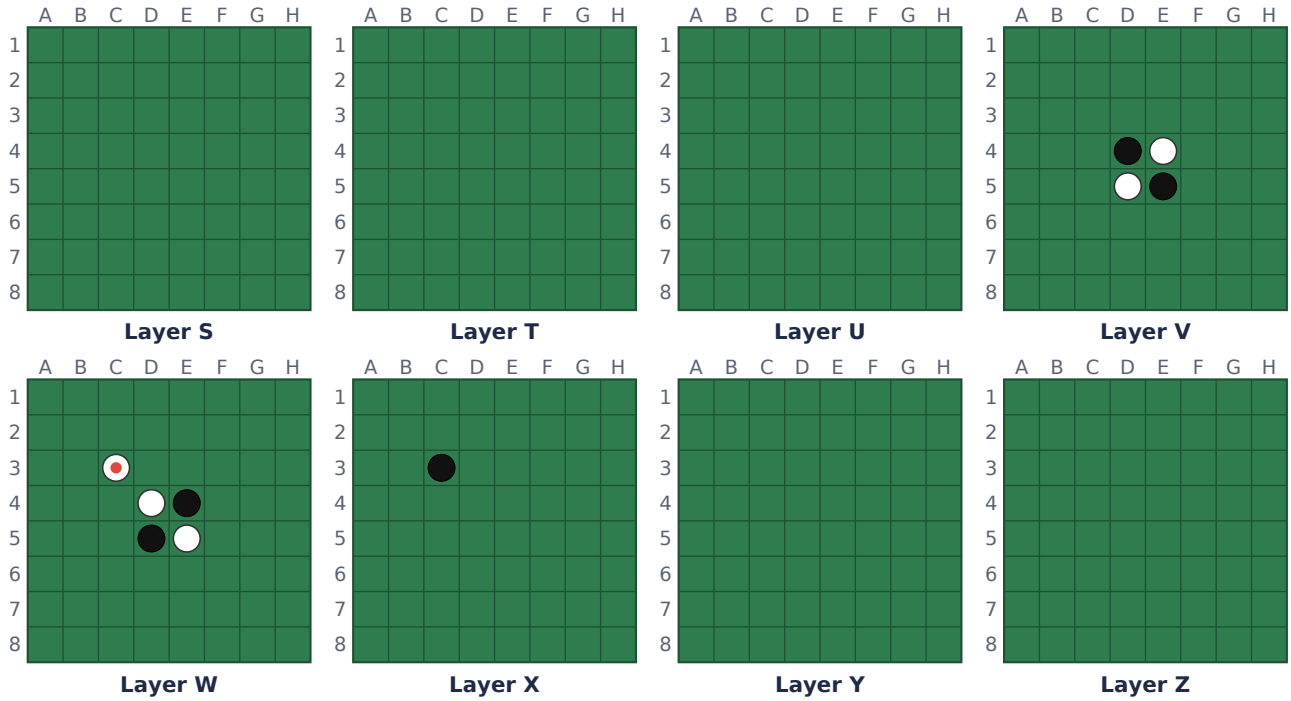


Black's best; White continues C-4-V.

Best replies (White): `C-4-V`, `D-5-X`, `E-6-V`

Quartz

`C-3-X C-3-W` (2 moves; discs 5–5; Black to move)



White plays right beside Black's disc. Black answers D-4-X or C-4-W.

Best replies (Black): `D-4-X`, `C-4-W`, `D-6-V`

Research and Acknowledgements

The development of Cube Reversi and its computer opponent has been informed by publicly available Othello/Reversi software, documentation, strategy resources, and research.

Research references include Othello-Sensei, Deft Reversi, ccurro/othelloAI, Otto Othello, the British Othello Federation, Othello Belgium, and other public Othello/Reversi resources.

Classic opening names were also checked against the weltyc/othello openings list, Edax's opening.c list, and the French Federation of Othello opening page.

Peter Alfeld's 3D Reversi pages (University of Utah) were a reference for Reversi played on a cube.

These projects and resources were consulted as research and design references. The Cube Reversi implementation is independently written.

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